

PHYSICS GRADE 10: MOMENTS OF EQUILIBRIUM

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.5 (6 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.5: Moments and Equilibrium
Total Duration	6 lessons × 40 minutes = 240 minutes total
Sub-Strand Content	– Turning effect of a force (moment of a force) – The Principle of Moments – Conditions for equilibrium of a body under parallel forces – Centre of gravity and its determination for regular and irregular objects – Stability and states of equilibrium (stable, unstable, and neutral) – Factors affecting stability: base area and position of centre of gravity – Application of moments and equilibrium in everyday life (see-saws, cranes, human body posture, Maasai warriors carrying loads)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - define the moment of a force and calculate moments in given situations, - state and apply the Principle of Moments to solve problems involving balanced systems, - state the conditions necessary for a body to be in equilibrium under the action of parallel forces, - determine the centre of gravity of regular and irregular objects experimentally, - explain the three states of equilibrium (stable, unstable, and neutral) with relevant examples, - investigate how the base area and position of the centre of gravity affect the stability of objects, - apply the concepts of moments and equilibrium to explain real-life situations such as the design of cranes, double-decker buses, and human body mechanics, - appreciate the role of moments and equilibrium in the safe design of structures and everyday objects.
Core Competencies	– Communication and Collaboration: the learner engages in group investigations (e.g., balancing metre rules, locating centre of gravity of cut-out shapes) and presents findings to the class, practising respectful scientific discourse and shared decision-making. – Critical Thinking and Problem Solving: the learner analyses why the Nyayo Bus topples on a slope or why a Maasai warrior adjusts posture when carrying a load, constructing logical arguments linking centre of gravity, base area, and moment calculations. – Learning to Learn: the learner reflects on predictions made before experiments (predict-observe-explain cycle) and revises mental models of balance and stability as new evidence emerges across all six lessons. – Digital Literacy: the learner uses digital simulations or video clips (e.g., slow-motion footage of the Fundi balancing a heavy

	roof truss) to gather evidence that would be impractical to obtain in the classroom, interpreting digital data to support scientific explanations.
Core Values	– Responsibility: the learner takes assigned roles during group experiments (measurer, recorder, spokesperson) and ensures equipment is handled safely and returned in good order, reflecting personal accountability. – Integrity: the learner records experimental results honestly, including unexpected data, and does not alter readings to match predictions, upholding scientific honesty. – Respect: the learner listens to and builds on the contributions of peers from different backgrounds during class discussions on how communities in Kenya (e.g., tea pickers in Kericho, matatu touts loading cargo) intuitively apply moments in daily work. – Unity: the learner collaborates across gender and ability lines during practical activities, recognising that safe, balanced structures benefit the whole community.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: students generate and post questions on the Driving Question Board after observing the anchoring phenomenon (a Jiko stool toppling vs. standing), framing what they need to investigate about forces, balance, and stability. – SEP 2 – Developing and Using Models: students build and iteratively revise a diagrammatic model (free-body / moment-arm diagram) across all six lessons, adding centre-of-gravity markers, moment arrows, and stability annotations as understanding deepens. – SEP 3 – Planning and Carrying Out Investigations: students design the procedure for locating the centre of gravity of irregular cardboard cut-outs and for verifying the Principle of Moments using a metre rule, masses, and a pivot — deciding variables, controls, and data-collection methods. – SEP 4 – Analysing and Interpreting Data: students calculate clockwise and anti-clockwise moments from tabulated experimental data, identify patterns, and evaluate whether their results support or contradict the Principle of Moments. – SEP 6 – Constructing Explanations: students use moment calculations and centre-of-gravity diagrams to explain why a loaded mkokoteni (handcart) tips over when the load is placed too far forward, connecting mathematical evidence to real-world observations. – SEP 8 – Obtaining, Evaluating, and Communicating Information: students read case-study cards on crane design and tower-crane accidents in Nairobi construction sites, evaluate the reliability of the sources, and communicate findings through annotated diagrams and oral presentations.
Pertinent & Contemporary Issues (PCIs)	– Safety and Accident Prevention: learners connect the Principle of Moments to vehicle safety — exploring why overloaded matatus and top-heavy lorries on Kenyan highways are prone to rolling, reinforcing road-safety consciousness. – Environmental Education: learners examine how improper loading of fishing boats on Lake Victoria shifts the centre of gravity dangerously, linking physics to the real environmental and human cost of capsizing accidents. – Gender Disparity: learners challenge stereotypes by studying contributions of women engineers and physicists (e.g., women working in Nairobi's construction and design sector) who apply moment principles daily, broadening perceptions of who does physics. – Life Skills — Critical Thinking and Problem Solving: learners evaluate everyday scenarios (carrying water jerricans, stacking sacks of maize) and propose physics-informed improvements, practising decision-making skills transferable beyond the classroom.
Career Connections	– Civil and Structural Engineer: uses moment calculations to design safe beams, bridges (e.g., the Thika Superhighway interchange), and building foundations that remain in equilibrium under load. – Mechanical Engineer / Crane Operator: applies the Principle of Moments to determine safe lifting loads and boom angles for tower cranes used on Nairobi high-rise construction sites. – Orthopaedic Physiotherapist / Sports Scientist: analyses how the human body's centre of gravity shifts during running

	(relevant to Kenyan elite marathon athletes) or injury rehabilitation, using moment principles to design corrective exercises. – Automotive / Transport Engineer: designs vehicle suspension and load-distribution systems to keep centre of gravity low and prevent rollover — directly relevant to matatu and bus safety standards in Kenya. – Aviation Technician / Pilot: applies equilibrium and moment principles to aircraft weight-and-balance calculations, a safety-critical skill for pilots operating from Wilson Airport, Nairobi. – Industrial / Ergonomics Designer: optimises the design of tools, furniture, and farm equipment (e.g., ox-plough balance) to reduce worker strain by aligning forces with comfortable moment arms.
Focus for Lessons	This unit investigates how and why objects balance, tip, or remain stable — anchored in the everyday Kenyan experience of objects toppling or staying upright. Students build understanding progressively: from the turning effect of a single force, through the Principle of Moments and conditions for equilibrium, to the role of centre of gravity in determining stability states. A phenomena-driven, investigation-centred approach is used throughout, with students revising a cumulative diagrammatic model in every lesson so that mathematical concepts ($\text{moment} = F \times d$) are always connected to visible, meaningful physical behaviour.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some objects stay balanced while others topple — and how can we use the physics of turning forces to predict and control whether an object will be stable or fall? KICD KEY INQUIRY QUESTIONS: 1. How is Physics relevant in day-to-day life? 2. How do forces determine the stability and equilibrium of objects we encounter every day?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Toppling Jiko Stool" A three-legged wooden Jiko stool (commonly found in Kenyan homes and market stalls) stands perfectly steady on a flat floor even when a heavy sufuria of ugali is placed near its centre. Yet when a student places a full 20-litre jerrycan of water on one corner of the stool and gently nudges it, the stool topples dramatically. Strikingly, a wider, lower four-legged stool with the same jerrycan placed at the same corner barely tilts — it simply rocks and returns to its upright position. Students observe that the height of the load, the width of the base, and where the load sits all seem to matter — but they cannot yet explain WHY in precise physical terms. This is puzzling because the same force (the weight of the jerrycan) causes toppling in one case and stability in another, suggesting that something about the geometry of the object — not just the force — controls whether it falls. The phenomenon hooks students into wanting to understand turning effects of forces, centre of gravity, and the conditions for equilibrium.
Storyline Thread	Lesson 1 — "Why Did the Stool Fall?" (Moment of a Force) Students observe the anchoring phenomenon (Jiko stool toppling vs. staying upright) and record initial predictions. They are introduced to the concept of the turning effect of a force — the moment — and derive the formula $\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$. They calculate moments for simple scenarios (e.g., pushing open a school gate at different distances from the hinge) and post their first "I notice / I wonder" questions on the Driving Question Board. The cumulative model begins: a diagram of the stool with force arrows. Lesson 2 — "What Makes a See-Saw Balance?" (Principle of Moments) Students investigate the Principle of Moments experimentally using a metre rule, a pencil pivot, and improvised masses (bags of beans, coins). They tabulate clockwise and anti-clockwise moments, identify the pattern, and state the Principle. They apply it to solve quantitative problems including a traditional Kenyan see-saw (bao plank across a log). The cumulative model gains moment-arm annotations and a balance condition label.

	Lesson 3 — "Where Is the 'Balance Point' Hiding?" (Centre of Gravity — Regular and Irregular Objects) Students determine the centre of gravity of regular shapes (ruler, rectangular cardboard) by calculation and of irregular shapes (cut-out map of Kenya, oddly shaped plywood) by the plumb-line suspension method. They link the position of centre of gravity to the balance-point concept established in Lesson 2. The cumulative model gains a centre-of-gravity marker (x) on the stool diagram. Lesson 4 — "Stable, Unstable, or Neutral — How Do We Decide?" (States of Equilibrium) Students test objects of different shapes and base widths (a tall thin bottle, a wide flat tin, a ball, a cone) by tilting them to small angles and observing whether they return, continue to fall, or stay tilted. They classify each as stable, unstable, or neutral and construct explanations linking each state to the vertical line through the centre of gravity and the base of support. The anchoring phenomenon is revisited: students now explain the stool's behaviour in these terms. The cumulative model gains stability-state annotations. Lesson 5 — "How Do Engineers Keep Things From Falling?" (Factors Affecting Stability and Real-Life Applications) Students investigate how lowering the centre of gravity (adding clay to the base of a tall cardboard box) and widening the base (adjusting cardboard flaps) each independently increase stability. They connect findings to real engineering contexts: crane design, double-decker bus geometry, and why maize sacks are stacked in an interlocking pattern. A case-study card on a Nairobi construction-site crane accident is analysed using moment and stability principles. The cumulative model is updated with stability-factor labels and an engineering example inset. Lesson 6 — "Can We Explain and Predict Toppling?" (Consolidation, Model Finalisation, and DQB Review) Students return to the full anchoring phenomenon and all supporting phenomena, using their completed cumulative model to write a full scientific explanation of when and why objects topple or remain stable. The Driving Question Board is reviewed: each posted question is answered with evidence from the unit. Students present their models in groups, peer-assess using a co-constructed rubric, and reflect on career connections. A summative problem set (including a jerrycan-on-stool moment calculation) is completed individually to assess mastery of learning outcomes.
Supporting Phenomena	– The Leaning Sugarcane Stack: At a Kisumu market, tall bundles of sugarcane leaning at a steep angle suddenly crash over while shorter, wider bundles remain stable — prompting questions about how the height of the centre of gravity relative to the base governs toppling. – The Balanced Metre Rule: A uniform metre rule balanced on a pencil pivot shifts dramatically when a 50 g mass (a small stone) is moved just 5 cm further from the pivot — a precise, classroom-observable event that makes the idea of a "turning effect depending on distance" concrete and measurable. – The Swaying Double-Decker Bus: A KBS double-decker bus navigating a sharp roundabout in Nairobi visibly leans outward; students observe (via video clip) that it does not tip — raising questions about how engineers keep the centre of gravity low enough for the bus to remain in equilibrium through the turn. – The Tightrope Acrobat's Pole: A circus acrobat at a Nairobi cultural festival carries a long, heavy pole held low and horizontal while walking a wire — counterintuitively, the pole makes balance easier, not harder, linking to the idea of deliberately lowering and widening the effective centre-of-gravity system to increase stability.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes): Why Did the Stool Fall? — Introducing the Moment of a Force

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon and introduce the concept of the moment of a force as a turning effect, anchoring all future learning in the question of why the Jiko stool topples.
Knowledge	By the end of this lesson, the learner should be able to: (a) define the moment of a force as the product of the force and the perpendicular distance from the pivot; (b) identify the pivot, force, and perpendicular distance in everyday Kenyan scenarios; (c) state that the moment of a force is measured in Newton-metres (N·m).
Skills	By the end of this lesson, the learner should be able to: (a) measure and record force and distance values to calculate moments; (b) use a simple beam-balance apparatus to observe turning effects; (c) construct an initial annotated sketch-model of the stool phenomenon showing force and pivot.
Attitudes	The learner appreciates that Physics principles — particularly turning effects of forces — are embedded in everyday Kenyan objects such as the Jiko stool, sufuria, and jerrycan, and begins to develop curiosity about predicting stability.
Key Inquiry Question	How do forces determine the stability and equilibrium of objects we encounter every day?
Purpose in Storyline	This is Lesson 1 of the Moments of Equilibrium storyline. It launches the anchoring phenomenon — the toppling Jiko stool — and opens the Driving Question Board. Students make their first predictions about why the stool topples, gather initial observational evidence about turning effects, and build a preliminary model. Every subsequent lesson will return to this stool scenario to deepen the explanation.
Safety Notes	Ensure the demonstration stool is placed on a stable, clear surface away from the edges of desks. The 20-litre jerrycan used in demonstration must be handled by the teacher only — not lifted by learners. Warn students to stand back (at least 1 metre) during the toppling demonstration. No glass apparatus in this lesson. Learners handling metre rules and masses should be reminded not to drop masses on feet.

B. LESSON OVERVIEW

Students encounter the anchoring phenomenon for the first time: a three-legged Jiko stool that stands firm under a sufuria of ugali but topples when a full 20-litre jerrycan is placed at its corner. Through structured prediction, a live teacher demonstration, a hands-on beam-balance activity, and guided sensemaking, students discover that forces create turning effects that depend on BOTH the size of the force AND the distance from a pivot. The lesson closes by opening the class Driving Question Board (DQB) with student-generated questions and an initial sketch-model of the stool phenomenon. No full explanation is reached — the mystery deepens purposefully.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to	Students enter to find a three-legged Jiko stool placed at the front of the room with a sufuria	1. Before class: set up the Jiko stool at the front. Place the empty sufuria on it. Write on the board:	Think-Pair-Share with Prediction Slip — this strategy activates prior knowledge about forces and	Collect Prediction Slips at the end of the phase. Observable indicators: (a) student has drawn

torque Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Torque (Flexbook) Source: CK-12 Search ARES for similar readings	(empty) resting near its centre. Without any introduction, they are handed a half-page 'Prediction Slip' and asked to silently read two scenario cards: – CARD A: 'A full sufuria of ugali (3 kg) is placed at the centre of the stool. Does it topple?' – CARD B: 'A full 20-litre jerrycan (20 kg) is placed on one corner of the same stool. A student gives it a gentle sideways nudge. Does it topple?' Students write their prediction (YES / NO / MAYBE), draw a quick force-arrow sketch on the stool diagram provided, and write ONE reason for their prediction. They then share predictions with a shoulder partner (Think-Pair) before three pairs share aloud to the class.	'PHENOMENON: Why did the stool fall?' Do NOT explain anything yet. 2. Distribute Prediction Slips as students enter. Say verbatim: 'Do not talk yet — read both cards and write your prediction silently. You have 3 minutes.' 3. After 3 minutes, say: 'Now turn to your shoulder partner — you have 2 minutes. Share your prediction and your reason. Try to agree or find where you disagree.' 4. After pair discussion, cold-call exactly 3 pairs using a random name stick. For each pair ask: 'What did you predict — and what was your reason?' [WAIT TIME: 10 seconds after each question before accepting an answer.] 5. Record the three predictions on the board in a T-chart labelled 'TOPPLES / DOES NOT TOPPLE'. Do NOT affirm or correct any prediction. Say: 'Hold these ideas — we will come back to test them very soon.' 6. Ask the whole class: 'What would you NEED to know or measure to be sure of your prediction?' Cold-call 2 students. [WAIT TIME: 12 seconds.] Write their responses under the T-chart as 'We need to find out...'	balance (from Grade 8 science), reveals existing misconceptions (e.g. 'heavier objects always topple'), and creates cognitive dissonance that motivates the rest of the lesson. Recording predictions publicly ensures accountability and allows the teacher to track conceptual change across the lesson.	at least one force arrow on the stool diagram; (b) student has given a reason that references either the weight, the position of the load, or the shape of the stool — even if imprecise. Flag slips where the student has given no reason or written only 'I don't know' for targeted scaffolding during Phase 3.
Observe Phase VIDEO: Introduction to torque - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos HTML:	PART A — Live Demonstration (10 min): The teacher performs a two-part live demonstration at the front. – Demo 1: Places the empty sufuria at the centre of the Jiko stool — stool stays upright. Places it near one edge — stool leans but does not fall. – Demo 2: Replaces the sufuria with a filled 20-litre jerrycan	PART A: 1. Before Demo 2, say verbatim: 'Watch carefully — I want you to notice exactly WHEN the stool starts to move and WHERE the jerrycan is sitting.' [Place jerrycan, step back, give a single gentle push.] 2. After the stool topples, say: 'Turn to your table group — 60 seconds — tell each other: what	Structured Observation Table + Hands-On Discovery — the two-part observe phase separates macroscopic phenomenon observation (stool demonstration) from quantitative pattern-finding (beam-balance). This dual-encoding approach ensures students connect the dramatic visual phenomenon to measurable variables before any formula is introduced. The beam-balance	Circulate and check Observation Tables during Part A. Observable indicators: (a) student correctly records at least 3 rows in the observation table; (b) student distinguishes between 'where the load is placed' and 'how heavy the load is' as separate variables. During Part B, listen for groups saying words like 'the further out, the more turning' — this indicates readiness for the moment formula

Torque Source: PhET Search ARES for similar readings	placed on one corner of the stool, then gives the stool a gentle nudge — the stool topples dramatically. – Demo 3: Moves the jerrycan to the wider, lower four-legged stool. Nudges it — the stool rocks and self-rights. Students complete an Observation Table in their notebooks with columns: OBJECT WHERE LOAD IS PLACED HEIGHT OF STOOL NUMBER OF LEGS RESULT (topple / stable). PART B — Beam-Balance Turning Activity (10 min): Each group of 4 receives: 1 × 30 cm wooden ruler balanced on a pencil (acting as a pivot at the centre), 3 × 50g masses, 1 × metre rule for measuring distances. Task: Place one 50g mass at 5 cm from the pivot on the LEFT side. Observe. Now place a second 50g mass at 10 cm from the pivot on the RIGHT side. Observe. Record what happens in each case. Groups then try to find a position for a single 50g mass on the right that balances a 100g mass placed 6 cm to the left. Record distances and masses used.	EXACTLY caused it to fall? Was it just the weight? Or was something else involved?' [WAIT TIME: allow full 60 seconds of group talk.] 3. Cold-call 2 groups for their observation (NOT explanation yet). [WAIT TIME: 10 seconds.] 4. Say: 'Write your observations in your table — not your explanation yet. Observations only.' PART B: 1. Circulate to each group during the beam-balance activity. At each group, ask: 'Which side is turning? What TWO things are you changing when you move the mass further out?' [WAIT TIME: 10 seconds at each group.] 2. If a group finds a balance position, ask: 'Can you see any pattern between the numbers on each side? Multiply the mass by the distance — what do you get on each side?' [Do NOT give the formula — let groups discover it.] 3. With 2 minutes remaining, say: 'Freeze. Write down in one sentence what you think the TURNING EFFECT depends on.'	activity embeds NGSS Science & Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data).	in Phase 3. Flag groups that only record 'it balanced' without recording the distances.
Explain Phase  VIDEO: Introduction to torque Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Torque (Flexbook) Source: CK-12	PART A — Concept Introduction via Guided Notes (8 min): Students open their ARES platform Guided Notes page for Lesson 1. The teacher leads a structured explanation: – Define MOMENT OF A FORCE: 'A moment is the turning effect of a force about a pivot point.' – Formula: Moment (M) = Force (F) × Perpendicular distance	PART A: 1. Write 'MOMENT = TURNING EFFECT' in large letters on the board BEFORE defining it formally. Say: 'You have all just observed moments — you just did not have the name for it yet.' 2. Write the formula slowly: $M = F \times d$. Say verbatim: 'The moment is NOT just about how heavy the object is — it also depends on HOW FAR from the pivot the force	Concept-Before-Formula anchoring with Phenomenon Callback — by naming the concept ('turning effect') before writing the formula, and immediately connecting the formula back to the stool demonstration, students encode $M = F \times d$ as a physical meaning rather than an abstract equation. The apply step (Part C) uses NGSS Practice 5 (Using Mathematics and Computational Thinking) to consolidate	Collect notebooks at the end of Part C (or photograph group work on ARES). Observable indicators: (a) student correctly applies $M = F \times d$ with correct units in the worked example; (b) student's beam-balance calculation shows moments within ± 1 N·m on both sides for the balanced case; (c) student's one-sentence conclusion references both force AND distance. Students who calculate only force OR only distance

Search ARES for similar readings	from pivot (d) → $M = F \times d$ – Unit: Newton-metre (N·m) – Students label a diagram showing: the pivot, the line of action of force, and the perpendicular distance — using the Jiko stool as the diagram context. PART B — Worked Example with Kenyan Context (5 min): Teacher works through: 'A sufuria containing 3 kg of ugali (weight = 30 N) is placed 0.2 m from the leg of a Jiko stool acting as the pivot. Calculate the moment of the force about that leg.' $M = F \times d = 30 \text{ N} \times 0.2 \text{ m} = 6 \text{ N}\cdot\text{m}$ PART C — Apply: Return to Beam-Balance Data (7 min): Groups retrieve their beam-balance data from Phase 2. They now calculate the moment on each side using $M = F \times d$ (using weight = mass $\times$ 10 N/kg). They check: when balanced, are the moments equal on both sides? Students write a one-sentence conclusion in their notebooks.	acts. This is why the jerrycan caused toppling even though the stool could hold the sufuria.' 3. Point back to the stool at the front: 'Where was the pivot in our demonstration?' Cold-call 1 student. [WAIT TIME: 10 seconds.] Confirm: the contact legs of the stool act as the pivot line. PART B: 1. Write the worked example on the board step by step. After writing $M = 30 \times 0.2$, pause and say: 'Before I multiply — tell your partner: will this moment be bigger or smaller if I move the sufuria to 0.4 m from the pivot? Discuss — 30 seconds.' Cold-call 1 student. [WAIT TIME: 10 seconds.] 2. Complete the calculation. Say: '6 Newton-metres. Write that unit carefully — it is NOT Newtons alone, and it is NOT metres alone. It is BOTH together.' PART C: 1. Say: 'Go back to your beam-balance data. You now have the tool to check whether the moments balanced. Do the calculation — 5 minutes.' 2. Circulate. If a group's moments do not match (due to measurement error), say: 'Good — science includes measurement uncertainty. What could have caused the difference?' [WAIT TIME: 10 seconds.] 3. Cold-call 2 groups to share their conclusions. Ask: 'Does your conclusion explain why the stool toppled? What was the larger moment in the stool demonstration?'	understanding through their own data.	without multiplying need immediate re-teaching of the formula meaning.
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Driving Question Board (DQB) Creation  VIDEO: Introduction to torque - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Torque Source: PhET  Search ARES for similar readings	Each student receives two sticky notes (different colours — e.g. yellow for 'I wonder...' questions, pink for 'I now understand...' statements). Students independently write: – YELLOW sticky: One question they still have about the stool phenomenon OR about moments/stability (e.g. 'Why does a wider base make it harder to topple?', 'Does the height of the jerrycan matter?', 'What is centre of gravity?'). – PINK sticky: One thing they now understand that they did NOT understand at the start of the lesson. Students walk to the classroom DQB (a large sheet of paper or whiteboard section) and place their stickies. The teacher facilitates a 3-minute gallery reading — students silently read the board. The teacher then clusters the yellow questions into 3–4 theme groups in front of the class and labels each cluster (e.g. 'Cluster 1: Where is the balance point?', 'Cluster 2: How does height matter?', 'Cluster 3: What makes something stable?'). These clusters will be revisited at the start of every subsequent lesson.	1. Distribute sticky notes. Say verbatim: 'The yellow note is for questions you STILL HAVE — things the stool phenomenon has made you wonder about. The pink note is one thing you now understand. You have 3 minutes — write independently, not with your partner.' 2. Invite students to the DQB one row at a time. Say: 'Place your stickies anywhere on the board — we will organise them together.' 3. After gallery walk, say: 'I can see some of your questions are about the same big ideas. Watch as I group them.' Physically move stickies into clusters in front of students, narrating: 'These three questions all seem to be about WHERE forces act — I'll call this cluster 'Location of Force.' These are about something called centre of gravity — we will get there in Lesson 3.' 4. Point to the clusters and say: 'This board is our science story. By the time we finish Sub-Strand 1.5, every cluster on this board should have an answer. Your job — and my job — is to fill this board with understanding, not just facts.' 5. Photograph the DQB for the ARES platform record. Cold-call 1 student: 'Which cluster do you most want to answer first — and why?' [WAIT TIME: 12 seconds.]	Driving Question Board (DQB) with Question Clustering — the DQB is a metacognitive tool (NGSS Practice 8: Obtaining, Evaluating, and Communicating Information) that externalises student thinking, tracks conceptual growth across the unit, and gives students ownership of the inquiry storyline. Clustering questions publicly shows students that science is organised around big ideas, not isolated facts.	Review all yellow sticky notes after class. Observable indicators: (a) at least 70% of students have written a genuine science question (not 'Will we do an experiment?'); (b) questions referencing 'centre of gravity', 'base width', 'height of load', or 'balance point' signal that Phase 2 successfully surfaced the key variables; (c) pink notes where students write 'I now understand moments = force × distance' confirm Phase 3 success. Collect photos of DQB as a class-level formative artefact.
Model Building Phase  VIDEO: Finding torque for angled forces <i>Related topic: not an exact match for this lesson.</i>	Students open a blank 'Model Builder' page (ARES platform or exercise book). Using the stool demonstration as context, each student draws an initial annotated model of WHY the stool toppled. The model must include: 1. A side-view sketch of the Jiko stool with the jerrycan on one	1. Say verbatim: 'Scientists build models to explain what they observe. Your model today does NOT need to be perfect — it needs to show your BEST THINKING right now. We will improve it in every lesson.' 2. Write the model requirements on the board as a checklist (items	Initial Model Construction with Group Comparison — initial model building (NGSS Practice 2: Developing and Using Models) makes student thinking visible and creates a baseline artefact for tracking conceptual change. Group comparison surfaces productive disagreements (e.g. different	Collect or photograph models before students leave. Observable indicators: (a) student has correctly placed the force arrow acting downward at the position of the jerrycan (not at the centre of the stool); (b) student has identified a pivot leg — even if not yet the correct one — showing

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Torque Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	corner. 2. A labelled downward force arrow (Weight of jerrycan, $F = 200\text{ N}$) at the position of the jerrycan. 3. A labelled pivot point (the contact leg on the opposite side). 4. A labelled perpendicular distance (d) from the pivot to the line of action of the force. 5. A calculated moment value ($M = F \times d$ using estimated or given values). 6. One sentence: 'The stool toppled because...' Students share their model with their group. Groups identify ONE thing all their models have in common and ONE thing that differs. One spokesperson reports to the class. Teacher notes areas of disagreement on the board as 'Model Revision Points — revisit in Lesson 2'.	1–6 above). Say: 'You have 7 minutes to draw your model independently.' 3. After 7 minutes, say: 'Compare with your group — 3 minutes. Find one agreement and one disagreement in your models.' 4. Cold-call 2 group spokespeople. [WAIT TIME: 10 seconds each.] For each report, ask: 'What caused the disagreement in your group — was it about the position of the pivot, or the direction of the force, or something else?' 5. Write disagreements on the board under 'Model Revision Points'. Say: 'These disagreements are EXACTLY what science looks like. We will resolve each of these in the coming lessons. Do not erase your model — we will look back at it at the end of the sub-strand to see how your thinking has changed.' 6. Close the lesson by pointing back to the stool: 'We now know that the jerrycan created a moment — a turning effect — about the pivot leg. But we still cannot fully explain WHY the four-legged stool did not topple. What are we missing? That question will drive Lessons 2 and 3.'	students placing the pivot at different legs) that create the intellectual need for more precise understanding. Publicly recording 'Model Revision Points' institutionalises the idea that models evolve with evidence.	understanding that a pivot is needed for a moment; (c) student has written a moment calculation with correct units ($\text{N}\cdot\text{m}$); (d) the 'The stool toppled because...' sentence references both force and distance rather than weight alone. Students who show only a weight arrow with no pivot or distance label require re-engagement with the $M = F \times d$ concept in Lesson 2.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. PREDICTION PHASE: Did students' initial predictions reveal specific misconceptions (e.g. 'only heavy objects topple', 'all three-legged stools are unstable')? Which misconceptions were most common — and how will you address them explicitly in Lesson 2?
2. OBSERVATION PHASE: Did the beam-balance activity surface the pattern $M = F \times d$ independently, or did groups need significant prompting? If most groups needed prompting, consider adding a data table with a pre-filled 'Force $\times$ Distance' column in Lesson 2 activities.
3. EXPLAIN PHASE: Were students able to connect the abstract formula $M = F \times d$ back to the physical stool? Listen for language in Phase 3 cold-calls — students saying 'the jerrycan was far from the pivot' show strong conceptual linkage; students saying only 'the jerrycan was heavy' need the distance component reinforced.
4. DQB QUALITY: Did the DQB generate questions about centre of gravity, base width, or load height? If students' questions remained only about 'why does it fall', the

phenomenon may need a more explicit second demonstration in Lesson 2 highlighting the four-legged stool contrast.

5. MODEL BUILDING: What were the most common 'Model Revision Points' identified during Phase 5? Use these to sequence the focus of Lessons 2 and 3. For example, if most disagreements were about pivot location, make pivot identification the opening activity of Lesson 2.

6. PACING: Was 80 minutes sufficient? If Phase 2 (beam-balance) overran, consider pre-assembling the rulers and masses before class, or reducing the number of balance configurations from 3 to 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A three-legged Jiko stool with a sufuria of ugali at its centre remained stable, but toppled when a full 20-litre jerrycan was placed at one corner and nudged. A wider, lower four-legged stool with the same jerrycan simply rocked and self-righted. In the beam-balance activity, moving a mass further from the pivot caused greater turning, and two different masses could balance when placed at specific distances on opposite sides.
What did I learn?	A force acting at a distance from a pivot creates a MOMENT (turning effect). The moment of a force is calculated as $M = F \times d$ (force multiplied by perpendicular distance from the pivot) and is measured in Newton-metres (N·m). The jerrycan created a large moment about the stool's pivot leg because of its large weight AND its distance from that leg — not weight alone.
How does this explain the phenomenon?	The stool toppled because the jerrycan's weight (200 N) acting at a large perpendicular distance from the pivot leg created a large clockwise moment ($M = F \times d$) that was not balanced by any opposing moment. The four-legged stool did not topple under the same conditions — suggesting that something about the geometry (width of base, position of pivot legs) affects whether the turning effect causes permanent toppling or is resisted. This 'something' — involving centre of gravity and conditions for equilibrium — remains to be explained in Lessons 2 and 3.

LESSON 2 (80 minutes): Turning Forces: Introducing the Moment of a Force

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To introduce the concept of the moment (turning effect) of a force, derive the formula $\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$, and apply it to simple real-life Kenyan scenarios, connecting the turning effect back to why the Jiko stool topples.
Knowledge	Learners will know that: (a) a force applied at a distance from a pivot produces a turning effect called a moment; (b) the moment of a force is calculated as $\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$ from the pivot; (c) the SI unit of moment is the Newton-metre (N·m); (d) increasing either the force or the perpendicular distance increases the moment.
Skills	Learners will be able to: (a) measure and record perpendicular distances from a pivot in a physical setup; (b) calculate moments using the formula $M = F \times d$; (c) compare moments in different physical configurations; (d) represent turning forces with annotated force-arrow diagrams; (e) post structured 'I Notice / I Wonder' questions on the Driving Question Board.
Attitudes	Learners will appreciate that: (a) precise measurement and careful observation are essential to understanding physical phenomena; (b) Physics concepts such as moments are embedded in everyday Kenyan objects and tools; (c) collaborative discussion deepens individual understanding.
Key Inquiry Question	How do forces determine the stability and equilibrium of objects we encounter every day?
Purpose in Storyline	In Lesson 1 students observed the Jiko stool phenomenon and made initial predictions about why the stool topples. In this lesson (Lesson 2 of 6) students gather the first piece of explanatory evidence: the idea that a force's turning effect depends not only on its magnitude but on WHERE it is applied relative to the pivot. This directly begins to explain why placing the jerrycan at the corner (far from the centre) has a greater toppling effect than placing it near the middle. The concept of the moment is the foundational building block — all subsequent lessons on the principle of moments, centre of gravity, and equilibrium conditions will rest on this lesson's derivation and formula.
Safety Notes	Ensure metre rules and masses are handled carefully to avoid dropping on feet. If a heavy sufuria or jerrycan is used as a demonstration prop, place it on a stable surface and never drop it. Students must remain seated during the teacher demonstration of the tilting stool to avoid injury if the stool tips. Wet floors near demonstration tables should be dried immediately.

B. LESSON OVERVIEW

Students re-engage with the anchoring phenomenon (the toppling Jiko stool) by revisiting their Lesson 1 predictions. The teacher demonstrates that the same force (weight of a jerrycan) has a dramatically different toppling effect depending on where it is applied relative to the stool's supporting legs. Through a guided hands-on activity using a metre rule balanced on a pencil pivot, students discover that moving a 500 g mass further from the pivot makes it harder to balance — i.e., the turning effect grows with distance. The class derives the formula $\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$ ($M = F \times d$), calculates moments for Kenyan-context scenarios (school gate hinges, a mama pushing a mkokoteni cart), and posts structured 'I Notice / I Wonder' questions on the Driving Question Board. The lesson closes with students adding the first annotated layer — force arrows showing moments — to their cumulative model diagram of the Jiko stool.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Introduction to torque Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Torque (Flexbook) Source: CK-12  Search ARES for similar readings	Students open their exercise books to their Lesson 1 prediction diagrams (the Jiko stool with the jerrycan at the corner vs. the centre). In pairs, they discuss: 'Your Lesson 1 prediction said the stool topples because of [their reason]. Today we are going to test that idea. Revise or defend your prediction: does the POSITION of the jerrycan change how hard the stool is to topple, and WHY?' Each student individually writes a revised prediction in one sentence, then shares with their partner. Four cold-called pairs share aloud. The teacher records key prediction vocabulary on the board in two columns: FORCE-BASED reasons (e.g., 'the jerrycan is heavy') vs. POSITION-BASED reasons (e.g., 'it is placed too far from the middle'). Students are told: 'By the end of today you will be able to say exactly HOW position changes the toppling effect — using a number and a formula.'	1. Display the Lesson 1 phenomenon image (or re-place the stool and jerrycan prop on the demonstration bench). Say verbatim: 'Cast your minds back to our Jiko stool. Last lesson you predicted WHY it topples. I want you to revisit that prediction now — silently, in your book — for 90 seconds.' [WAIT 90 seconds — circulate, do NOT prompt.] 2. Say: 'Now discuss with your partner: does WHERE the jerrycan sits on the stool change the toppling effect? Give a reason.' [WAIT 2 minutes for pair talk.] 3. Cold-call exactly 4 pairs (choose 1 force-based, 1 position-based, 1 mixed, 1 uncertain). After each: 'Thank you. Class, do you agree or disagree? Thumbs up or down.' [WAIT 10 seconds for thumbs.] 4. Annotate the board columns. Say verbatim: 'Notice we have two types of ideas — force-based and position-based. Today we will find out which one Physics says is the key to the toppling puzzle — or whether it is BOTH.'	Strategy: PREDICTION REVISION (Elicit-Confront-Resolve). Students surface their prior mental models before instruction. Sorting predictions into 'force-based' vs. 'position-based' columns creates a visible cognitive conflict — students see that their class has two competing explanations. This tension motivates the evidence-gathering activity that follows and gives the teacher diagnostic insight into which misconceptions (e.g., 'it topples just because the jerrycan is heavy') need to be directly addressed.	OBSERVABLE INDICATOR: After the 4 cold-calls, at least 60% of students should be able to state that BOTH the magnitude of the force AND its position relative to the pivot seem to matter (thumbs up for position-based explanations). If fewer than half agree, the teacher pauses and restates: 'Look at the stool — if I put the same jerrycan right in the CENTRE, does it topple? [No.] But at the corner? [Yes.] Same weight — different position. Write that in your book as a KEY OBSERVATION.' This is a critical gate before advancing to Phase 2.
Observe Phase  VIDEO: Introduction to torque - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Torque Practice Source: CK-12  Search ARES for similar readings	Groups of 3–4 students receive: one metre rule, one pencil (acts as the pivot/fulcrum placed at the 50 cm mark), one 500 g mass, and a printed data table. Task: Place the 500 g mass at 10 cm from the pivot, then 20 cm, 30 cm, 40 cm, and 50 cm from the pivot. At each position, students observe whether the rule tips and record in the table: (a) the distance from pivot in cm, (b) a qualitative observation (balanced / tips slightly / tips strongly), and (c) a sketch showing the mass position and the direction the rule tips. After completing the table, each group answers on a	1. Distribute equipment. Say verbatim: 'You have 12 minutes. Your job is NOT to explain yet — just observe and record carefully. What changes as you move the mass further from the pivot? Fill in the table. Silence for the first 2 minutes so everyone makes their own first observation.' [WAIT 2 minutes — silent individual observation.] 2. After 2 minutes: 'You may now discuss quietly in your group as you continue the experiment.' 3. At 10 minutes: 'Two minutes left — write your pattern statement on your sticky note and post it on the	Strategy: STRUCTURED OBSERVATION WITH PATTERN DETECTION. Students generate empirical data (qualitative observations across 5 distance positions) before any formula is introduced. The sticky-note Evidence Wall makes all groups' patterns visible simultaneously, allowing students to see convergent findings across different groups — building confidence that the pattern is real and repeatable. The door demonstration bridges the abstract metre-rule result to a lived Kenyan school experience, reinforcing that	OBSERVABLE INDICATOR: Check the sticky notes on the Evidence Wall. Every group should state a pattern equivalent to: 'The further the mass from the pivot, the stronger the tipping effect.' If any group states only 'the mass made it tip' (ignoring distance), the teacher directs them: 'Look at your table — at 10 cm vs. 50 cm, does the mass change? No. What changes?' This is done publicly so all groups benefit. An additional formative check: ask all students to hold up their data tables — scan for complete entries across all 5 distance positions. Incomplete

	sticky note: 'What pattern do you notice between distance and tipping?' Groups stick their sticky notes on the class 'Evidence Wall.' The teacher then performs a brief dramatic demonstration: pushing open the school's gate (or a classroom door) near the hinge vs. near the outer edge, asking 'Which feels HARDER? Why?'	Evidence Wall.' 4. Perform the door/gate demonstration. Ask: 'Raise your hand if it felt harder to push near the hinge.' [WAIT 10 seconds.] Cold-call one student: 'Kwame, describe in your own words WHY you think it felt harder near the hinge.' 5. Say verbatim: 'This is the SAME idea as our metre rule experiment — and the SAME idea as our Jiko stool. The turning effect of a force changes with distance from the pivot. Physicists give this turning effect a name. Any guesses?' [WAIT 15 seconds — accept all guesses without affirming yet.]	this is not a textbook abstraction.	tables indicate a procedural gap that must be closed before Phase 3.
Explain Phase  VIDEO: Introduction to torque Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Torque (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher leads a whole-class derivation of the moment formula, anchored in the metre rule data. Students follow along in their books, filling in a guided notes frame (printed or dictated). Step 1 — Naming: Students learn the term 'moment of a force' (or 'torque') and its definition: the turning effect produced by a force about a pivot. Step 2 — Formula derivation: using the metre rule data, students observe that doubling the distance doubled the tipping effect → the relationship is proportional → $\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$, or $M = F \times d$. Step 3 — Units: Force in Newtons, distance in metres → moment in Newton-metres (N·m). Step 4 — Worked examples: (a) A mama applies a 40 N force to push a mkokoteni cart handle 0.8 m from the wheel axle pivot — calculate the moment. (b) A student pushes a school gate with 25 N at 1.2 m from the hinge — calculate the moment. (c) The	1. Write on the board: 'MOMENT OF A FORCE' in large letters. Say verbatim: 'This is the Physics name for the turning effect you just discovered. Write this definition word for word: The moment of a force is the product of the force and the perpendicular distance of its line of action from the pivot.' 2. Say: 'From your experiment — when you doubled the distance from 10 cm to 20 cm, what happened to the tipping effect?' [WAIT 10 seconds.] Cold-call 2 students. 3. Write on board: $M = F \times d$. Walk through the derivation verbally: 'Moment equals Force times perpendicular distance. This is our formula for today.' 4. Work Example (a) aloud on the board. Then say: 'Now YOU do Example (b) — solo — 3 minutes. No talking.' [WAIT 3 minutes.] Then: 'Compare your answer with your partner — 1 minute.' Cold-call one pair to share. Correct on board if needed.	Strategy: GUIDED DERIVATION WITH PHENOMENON ANCHORING. Rather than presenting $M = F \times d$ as a received formula, the teacher constructs it from the students' own experimental data, making the formula feel discovered rather than imposed. Each worked example uses a Kenyan context (mkokoteni cart, school gate) so the formula immediately feels applicable rather than abstract. The explicit 'EVIDENCE STATEMENT 1' step creates a written knowledge artifact that students will build on in each subsequent lesson — directly connecting new knowledge to the anchoring phenomenon.	OBSERVABLE INDICATOR 1: During solo work on Example (b), circulate and check that students are (i) converting cm to m, (ii) multiplying $F \times d$ (not adding), and (iii) writing the unit N·m. A common error is using cm instead of m — address this publicly: 'I am seeing some books with units of N·cm. Is that correct? What is the SI unit of distance?' [WAIT 10 seconds.] OBSERVABLE INDICATOR 2: The Jiko stool moment calculation should yield approximately $200 \text{ N} \times 0.35 \text{ m} \approx 70 \text{ N}\cdot\text{m}$ (depending on student estimates). Accept any reasonable estimate — the key formative check is whether students correctly identify 'd' as the distance from the jerry can to the nearest supporting leg (the pivot), not the distance to the centre of the stool. If students use the wrong distance, prompt: 'Where does the stool pivot when it begins to topple? Which leg acts as the fulcrum?'

	same student pushes with 25 N at only 0.3 m from the hinge — compare the moments and explain which is easier. Step 5 — Phenomenon reconnection: students look at their Jiko stool diagram and annotate: 'When the jerrycan is at the corner, d is LARGE → moment is LARGE → large turning effect → stool topples.'	5. Say: 'Now look at your Jiko stool diagram from Lesson 1. Where is the jerrycan placed when the stool topples? Use a ruler — estimate the distance from the nearest leg. Write the moment calculation — estimate the force as 200 N (20 kg jerrycan). What moment do you get?' [WAIT 3 minutes individual work.] 6. Verbatim closing statement: 'So the stool does not topple just because the jerrycan is heavy. It topples because the jerrycan is FAR from the pivot — the supporting leg — so the moment is LARGE. This is our first answer to the driving question. Write it in your book as EVIDENCE STATEMENT 1.'		
Driving Question Board (DQB) Creation  VIDEO: Introduction to torque - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Torque Practice Source: CK-12  Search ARES for similar readings	Each student individually writes TWO sticky notes: one 'I NOTICE...' and one 'I WONDER...' based on today's lesson AND the anchoring phenomenon. They then read their partner's sticky notes, agree on the best one from each category, and post them on the class DQB under the columns: 'WHAT WE NOW KNOW' and 'QUESTIONS WE STILL HAVE.' The teacher reads aloud 4–5 posted wonders, clusters similar questions, and explicitly labels emerging sub-questions (e.g., 'Several of you are asking: what happens when there are TWO forces turning in OPPOSITE directions? That is what we will investigate in Lesson 3.'). The teacher also reviews any sticky notes from Lesson 1 that today's lesson has now answered, moving them from 'Questions We Have' to 'What We Now Know,' modelling that the DQB is a living record of growing understanding.	1. Distribute two sticky notes per student. Say verbatim: 'You have 3 minutes — solo — to write one I NOTICE and one I WONDER. Your I NOTICE must use the word MOMENT. Your I WONDER must be a question you genuinely cannot yet answer.' [WAIT 3 minutes — no talking.] 2. 'Share with your partner — choose the best one from each category — 1 minute.' [WAIT 1 minute.] 3. 'Come post your chosen sticky notes.' [Allow 2 minutes for posting.] 4. Read 4–5 wonders aloud. Say: 'I am seeing a cluster here — three groups are asking about what happens when forces pull in OPPOSITE directions. Excellent — that is exactly where we are heading.' Write the cluster label on the DQB. 5. Move at least one Lesson 1 sticky note to 'What We Now	Strategy: DRIVING QUESTION BOARD CURATION. The DQB is not a decoration — it is a publicly visible epistemological record. By explicitly moving answered questions and labelling new clusters, the teacher models scientific knowledge-building as cumulative and self-correcting. Students see their own questions becoming the roadmap for future lessons, which increases ownership and motivation. The constraint ('Your I NOTICE must use the word MOMENT') ensures students encode the lesson's key term in their own language rather than copying a definition.	OBSERVABLE INDICATOR: Scan all posted 'I NOTICE' sticky notes before clustering. Every note should reference 'moment' and 'distance' or 'pivot.' Notes that say only 'forces cause turning' (without the distance relationship) indicate that the $M = F \times d$ derivation did not fully land. The teacher makes a mental note of the authors of vague notices and will target them in the opening retrieval activity of Lesson 3. 'I WONDER' notes that ask 'what if two forces act at the same time?' or 'what if forces act in opposite directions?' signal readiness for the principle of moments (Lesson 3) — celebrate these publicly as 'great scientific questions.'

		Know.' Say verbatim: 'In Lesson 1, Aisha asked: does WHERE you place the weight matter? Today's formula gives us our answer — yes, and we can now CALCULATE exactly how much it matters. That question is answered. It moves here.'		
Model Building Phase  VIDEO: Finding torque for angled forces <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Torque <i>Related topic: not an exact match for this lesson.</i> Source: PhET  Search ARES for similar readings	Students turn to their Lesson 1 cumulative model page (a diagram of the Jiko stool with the jerrycan). Using the knowledge from today's lesson, they add a NEW LAYER to the model: (1) Draw and label a force arrow representing the weight of the jerrycan, pointing downward, with its magnitude (200 N). (2) Draw and label a dashed line showing the perpendicular distance 'd' from the force's line of action to the nearest supporting leg (pivot). (3) Write the calculated moment value (in N·m) next to the arrow. (4) Add a caption: 'Large d → Large Moment → Large turning effect → Stool topples.' Students also add a contrasting mini-diagram: the same jerrycan placed at the centre of the stool — shorter d, smaller moment — with the caption 'Small d → Small Moment → Stool stays upright.' At the bottom of the model page, students write their current 'best explanation' of the driving question in 2–3 sentences, acknowledging what they still do not know.	1. Say verbatim: 'Open your Lesson 1 model diagram. You are NOT drawing a new diagram — you are ADDING to the same one. Scientists revise their models as they learn more. Today you are adding LAYER 2: moments.' Display a teacher model on the board (pre-drawn stool outline) and annotate live as you give instructions. 2. Step through each annotation aloud: 'First — force arrow on the jerrycan, pointing DOWN, labelled $W = 200\text{ N}$. Second — dashed line from the arrow to the nearest leg. Label it d. Third — write your moment calculation. Fourth — caption.' [WAIT 5 minutes for students to draw.] 3. Say: 'Now add the contrasting diagram — same stool, jerrycan in the CENTRE. Smaller d. What happens to the moment? Calculate it.' [WAIT 3 minutes.] 4. 'Final step — write your 2–3 sentence best explanation of our driving question using the words: moment, force, distance, pivot, topple.' [WAIT 3 minutes.] 5. Cold-call 2 students to read their explanation aloud. After each, ask the class: 'Does this explanation use evidence from our experiment today? Thumbs up or down.' [WAIT 10 seconds for thumbs.]	Strategy: CUMULATIVE MODEL REVISION (Layered Annotated Diagram). The model is the same diagram across all 6 lessons, with each lesson adding one explanatory layer. This structure makes conceptual growth visible and tangible — students can literally point to what they knew in Lesson 1 vs. what they know now. The contrasting mini-diagram (jerrycan at centre vs. corner) forces students to apply the formula comparatively, not just computationally, deepening conceptual understanding. The 'best explanation so far' paragraph activates metacognition: students must articulate both what they now know AND what they still cannot explain, sustaining inquiry momentum.	OBSERVABLE INDICATOR: Collect (or photograph) 6–8 model diagrams at random before students leave. Check for: (a) force arrow pointing downward with N label ✓; (b) dashed perpendicular distance line from arrow to pivot leg ✓; (c) correct $M = F \times d$ calculation with N·m unit ✓; (d) captions distinguishing large-d (topples) from small-d (stable) ✓. A model with a force arrow but NO distance annotation indicates the student can name the formula but has not yet connected 'distance from the pivot' to the physical geometry of the stool — this student needs a targeted question at the start of Lesson 3: 'Point to the pivot on your stool diagram. Now point to the distance d. Are they the same point?'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) **FORMULA VS. CONCEPT**: Did students calculate $M = F \times d$ correctly while still confusing which point is the pivot? If yes, design a 3-minute 'pivot identification' retrieval activity to open Lesson 3. (2) **PHENOMENON CONNECTION**: When students annotated the Jiko stool model, did they instinctively measure 'd' from the jerrycan to the nearest leg (correct pivot), or to the centre of the stool (common error)? The correct pivot is the leg that acts as the fulcrum during tipping — if students used the centre, address this explicitly at the start of Lesson 3. (3) **DQB QUALITY**: Count the number of 'I Wonder' questions that ask about TWO forces or OPPOSITE directions. If more than half the class generated such questions, you can introduce the Principle of Moments in Lesson 3 with confidence that students are cognitively ready. If fewer than a quarter did, begin Lesson 3 with a 5-minute re-engagement using the metre rule: 'What if I put a mass on BOTH sides of the pivot — can I make it balance? Try it.' (4) **ENGAGEMENT**: Did the mkokoteni and school-gate examples resonate as genuinely familiar? If students seemed unfamiliar with the mkokoteni, substitute a wheelbarrow used on a Kenyan farm, or a water pump handle at a borehole in the next lesson's examples. (5) **PACING**: If Phase 3 (Explain) ran over time, consider pre-printing the worked examples rather than dictating them, so instructional time focuses on the reasoning steps rather than copying. Note which groups completed the contrasting mini-diagram in Phase 5 — groups that did not reach it should be given it as a 5-minute starter task in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that moving a 500 g mass further from the pencil-pivot on a metre rule increased the tipping effect, even though the mass (and therefore the force) did not change. They also observed, in a teacher demonstration, that pushing a door near the hinge requires far more effort than pushing near the outer edge. In the anchoring phenomenon, they re-observed that placing the jerrycan at the corner of the Jiko stool (far from the supporting legs) produced a dramatic toppling effect that placing it near the centre did not.
What did I learn?	Students learned that the turning effect of a force is called its MOMENT , and it is calculated using the formula $M = F \times d$, where F is the applied force in Newtons and d is the perpendicular distance from the force's line of action to the pivot, in metres. The SI unit of moment is the Newton-metre (N·m). A larger force OR a larger perpendicular distance produces a larger moment. This explains why the jerrycan at the corner of the Jiko stool produces a larger toppling moment than the same jerrycan placed at the centre.
How does this explain the phenomenon?	Students can now partially explain the anchoring phenomenon: the Jiko stool topples when the jerrycan is at the corner because the perpendicular distance from the jerrycan's weight (force) to the nearest supporting leg (pivot) is large, producing a large moment (turning effect). When the jerrycan is near the centre, the distance d is small, so the moment is small and the stool remains upright. What students cannot yet explain: what happens when there are turning forces acting in OPPOSITE directions (clockwise and anticlockwise) — this is the Principle of Moments, which is the focus of Lesson 3.

LESSON 3 (80 minutes): Lesson 3: Centre of Gravity and the Tipping Point — Why Does the Jiko Stool Fall?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate how the position of the centre of gravity relative to the base of support determines whether an object is stable, unstable, or in neutral equilibrium.
Knowledge	By the end of the lesson, the learner should be able to: (a) define centre of gravity as the point through which the total weight of a body appears to act; (b) explain stable, unstable, and neutral equilibrium using centre of gravity and base of support; (c) state the conditions under which a body topples — that is, when the vertical line through the centre of gravity falls outside the base of support.
Skills	By the end of the lesson, the learner should be able to: (a) locate the centre of gravity of irregular flat objects using a plumb-line method; (b) design and carry out an investigation to compare the stability of objects with different base widths and heights; (c) construct evidence-based arguments linking centre of gravity height and base width to toppling behaviour observed in the Jiko stool phenomenon.
Attitudes	The learner appreciates that physical principles governing balance and stability are embedded in everyday Kenyan objects — from Jiko stools to water-carrying jerrycans — and values the role of careful observation and measurement in building scientific explanations.
Key Inquiry Question	How do forces determine the stability and equilibrium of objects we encounter every day?
Purpose in Storyline	In Lesson 2, learners used the principle of moments (sum of clockwise moments = sum of anticlockwise moments) to explain why a see-saw balances at a particular pivot position. Lesson 3 now advances the storyline by asking: given that turning forces matter, WHERE exactly does the total weight of the Jiko stool — and its load — act? Learners discover the centre of gravity concept, investigate how its height and the width of the base of support together determine the toppling condition, and build a precise geometric explanation for why the jerrycan-loaded stool falls. This sets up Lesson 4, which will quantify the restoring and toppling moments to predict the exact tipping angle.
Safety Notes	Ensure all stool/block models and jerrycan props are placed on the floor or a stable bench — never on a desk edge. Students must not stand on stools during demonstrations. When suspending objects from a plumb line, ensure the string is securely tied and scissors are handled safely. Floors must be clear of bags and water to prevent slipping if an object falls. Teacher supervises all group tilting activities and reminds learners to tilt objects slowly and in a controlled manner.

B. LESSON OVERVIEW

Learners begin by revisiting the anchoring phenomenon — the toppling Jiko stool — and the DQB from Lessons 1 and 2. The teacher introduces a new sub-question: 'But WHERE does the weight of the stool actually pull from?' Learners first predict where the centre of a flat cardboard shape 'balances', then discover the centre of gravity experimentally using a plumb-line method with irregular cardboard cut-outs shaped like a Jiko stool profile and a sufuria lid. Groups then investigate how raising the centre of gravity (stacking blocks or books) and narrowing the base affects when a model 'topples'. The Explain phase formalises the three states of equilibrium with diagrams and connects them directly to the stool-and-jerrycan scenario. The DQB is updated, and learners revise their cumulative explanatory models with annotated diagrams showing the centre of gravity, base of support, and the critical tipping line.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: moments, equilibrium centre, gravity  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: moments, equilibrium centre, gravity	Learners are shown a cardboard silhouette cut in the profile of a Jiko stool (wide base, narrow top) and an irregularly shaped piece of cardboard resembling a sufuria lid. Working individually, each learner marks with a pencil dot where they think the object's weight is 'centred' — the single point from which the whole weight seems to pull downward. Learners then share their predictions with a shoulder partner, noting agreements and disagreements. The teacher collects predictions on the board using a quick show-of-hands heat map (5 zones marked on a projected image of the silhouette). Learners also write a one-sentence prediction: 'I think the stool topples because ...', connecting to their prior knowledge from Lesson 2 on moments.	Teacher displays the two cardboard cut-outs (Jiko stool profile and sufuria-lid shape) using a document camera or by holding them up clearly. Teacher says: 'Before we touch any equipment, I want you to mark — just with a mental dot — where you think the entire weight of this shape is trying to pull downward from. You have 30 seconds to decide silently.' [WAIT 30 seconds — no talking.] Teacher then says: 'Turn to your shoulder partner. You have 60 seconds — tell each other WHERE you marked and WHY you placed it there.' [WAIT 60 seconds.] Teacher cold-calls 4 learners (one from each corner of the room): 'Amina — where did you mark it and what was your reasoning?' Repeats for Ochieng, Wanjiku, and Mwangi. Teacher records the four prediction zones on the board without affirming or correcting. Teacher then asks the class: 'We used the PRINCIPLE OF MOMENTS in Lesson 2. How might that idea help us find this balance point today?' [WAIT 10 seconds.] Cold-calls 2 learners. Teacher closes the Predict phase: 'Hold your prediction — we are about to find out who is right, and more importantly, WHY it matters for our toppling stool.'	Strategy: Prediction Mapping. By committing a prediction to paper and to a partner before any instruction, learners activate prior knowledge (Lesson 2 moments) and generate cognitive dissonance when their predictions inevitably scatter across different zones. This productive uncertainty primes learners to notice and explain the evidence they are about to gather, rather than passively receiving the answer.	Observable indicator: Teacher scans individual prediction dots and listens to partner-talk. Look for whether learners place the dot near the geometric centre vs. near the heavy base — a common misconception is that it must be at the exact geometric centre. If more than half the class places the dot at the geometric centre of the irregular shape (ignoring mass distribution), teacher notes this as a target misconception to address explicitly in the Explain phase.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms:	ACTIVITY A — Plumb-Line Centre of Gravity Hunt (12 min): Each group of 4 receives: one irregular cardboard cut-out (Jiko stool profile), one sufuria-lid shape, a pin-board or corkboard, a length of string with a small nut/bolt as a plumb bob, a ruler, and a pencil. Learners suspend each shape	Teacher circulates during Activity A. At the 4-minute mark, teacher pauses groups: 'Freeze. Hold up your first plumb line. What direction is the string pulling?' [WAIT 10 seconds.] 'That direction IS the direction gravity is pulling. Draw that line NOW.' Teacher models on one group's board if	Strategy: Evidence Sketching with Annotated Diagrams. Requiring learners to draw the vertical CG line on their tipping tower sketches forces them to spatially relate the abstract concept of centre of gravity to the observable physical event of toppling. This embodied, visual record becomes the primary	Observable indicator: In Activity A, all groups should find that their three plumb lines converge at a single point — if they do not, teacher checks that the shape was truly allowed to hang freely and the line was drawn while hanging (not after removal). In Activity B, the key observable indicator is

moments equilibrium centre gravity  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: moments equilibrium centre gravity	from three different pin-holes along its edge, each time hanging the plumb-line string from the same pin and drawing the vertical line it traces across the cardboard. The intersection of the three lines is labelled 'CG'. Groups compare their CG mark to their original prediction dot. ACTIVITY B — Tipping Towers (8 min): Each group receives a stack of 4 wooden blocks (or thick textbooks). Groups build two towers: Tower A = wide base (4 blocks side by side, 1 block high) and Tower B = narrow base (1 block wide, 4 blocks tall). Groups place a small 500 g mass (a bag of maize or a water bottle) first on the centre, then gradually shift it to the edge, observing at what point each tower topples. Groups record: (i) which tower toppled first, (ii) how far the mass had moved when toppling occurred, (iii) a sketch showing the tower, mass, and a dotted vertical line from the estimated CG.	needed. At the 8-minute mark, teacher asks the whole class (raised-voice whole-class prompt): 'Before you move to Activity B — where did your three lines meet? Is it inside the shape or outside? Thumbs up if inside, thumbs sideways if on the edge, thumbs down if outside.' Teacher notes the distribution. During Activity B, teacher circulates and asks targeted groups: 'Ochieng's group — when the mass was at the corner and the tower fell, describe exactly what happened to that vertical CG line just before it fell.' [WAIT 10 seconds.] Cold-calls 3 groups to share their tipping observations. Teacher does NOT give the explanation yet — instead says: 'Make sure your sketch captures EXACTLY where the CG was when it tipped. That sketch is your evidence.'	evidence artefact they will use in the Explain phase to construct a causal argument.	whether learner sketches show the CG vertical line moving beyond the base edge at the tipping point — if sketches do not show this, teacher prompts: 'Draw a straight vertical line downward from the CG mark. Does it land inside or outside the base?'
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: moments equilibrium centre gravity  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings	Groups share their Activity B sketches via a Gallery Walk (each group posts their sketch on the wall; learners circulate for 4 minutes with sticky notes, writing one observation per sketch). The class then reconvenes. Using their evidence, learners co-construct definitions of the three states of equilibrium — stable, unstable, and neutral — guided by teacher questioning. Learners draw three canonical diagrams in their notebooks: (1) a wide-based cone — CG low, vertical line falls inside base after tilt → returns; (2) a narrow tall block — CG high, vertical line falls outside base after small tilt → topples; (3) a sphere	Teacher opens the Gallery Walk debrief: 'I saw some fascinating sketches. Wanjiku's group — read out the sticky note someone left on your sketch.' After 2 responses, teacher says: 'Let's build our shared explanation together. Look at your Activity B sketches. In EVERY case where the tower fell — where was the vertical CG line?' [WAIT 12 seconds.] Cold-calls 3 learners. Teacher writes on the board: 'A body topples when the vertical line through its centre of gravity passes _____ the base of support.' Teacher waits — does NOT fill in the blank. [WAIT 10 seconds.] Cold-calls learner: 'Mwangi — finish that sentence.'	Strategy: Co-constructed Definition from Evidence (Claim-Evidence-Reasoning). Rather than presenting definitions first, the teacher uses the Gallery Walk and targeted questioning to pull the definition out of learners' own evidence sketches. This ensures that the formal vocabulary of 'centre of gravity' and 'base of support' is anchored to physical observations, making the concepts retrievable and transferable rather than memorised labels.	Observable indicator: Learner annotations of the Jiko stool diagram must show (a) the CG marked at a plausible location above the stool seat (higher when jerrycan is on top), (b) a vertical line drawn downward from CG, and (c) a written sentence stating whether the line falls inside or outside the base triangle. Teacher collects 6 annotated diagrams (one per group) as exit evidence. If learners cannot place the CG correctly — for example, placing it at the jerrycan itself rather than the combined system — teacher re-prompts: 'The jerrycan and the stool are ONE system now. Where is the combined weight acting?'

Search terms: moments equilibrium centre gravity	on a flat surface — CG always above contact point → rolls without toppling. Learners then apply the model directly to the anchoring phenomenon: they annotate a printed diagram of the Jiko stool (one with the jerrycan at the centre, one with it at the corner) showing: CG position, base of support, and the vertical CG line, then write a two-sentence causal explanation of why the stool topples in one case and not the other.	Teacher formalises the vocabulary: 'Centre of gravity', 'base of support', 'stable equilibrium', 'unstable equilibrium', 'neutral equilibrium' — writing each term as learners define them from evidence, not the other way around. Teacher then directs: 'Now apply this to OUR stool. On the diagram in front of you — draw the CG for the jerrycan-at-corner case. Draw the vertical CG line. Does it fall inside or outside the three-legged base triangle? That is your answer to why the stool topples.' Teacher circulates for 3 minutes while learners annotate. Teacher cold-calls one group to explain their diagram to the class. Teacher closes: 'This is the physics our Kenyan jua kali craftspeople use — even if they have never written it down — when they decide how wide to make a stool's legs.'		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: moments equilibrium centre gravity  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms:	The class reviews the DQB constructed in Lessons 1 and 2. Each group selects two sticky notes from the DQB that they believe Lesson 3 has now given them evidence to answer or partially answer. Groups write a revised answer card (green sticky note) and place it beside the original question (yellow sticky note). The class also generates at least two NEW questions provoked by today's lesson — for example: 'Can we calculate EXACTLY how far we can move the jerrycan before the stool tips?' or 'Does the shape of the base (triangle vs. square) change how stable the stool is?' — and adds these to the DQB as orange sticky notes. The teacher records any questions that directly anticipate Lesson 4 content (calculating toppling	Teacher points to the DQB: 'We started this unit with big questions about the toppling stool. Let's see what we can NOW answer with today's evidence.' Teacher reads aloud two questions from the DQB: 'Why does the stool topple when the jerrycan is at the corner?' and 'Does the height of the load matter?' Teacher says: 'Discuss with your group: can we answer either of these NOW? Use your diagrams as evidence. 90 seconds.' [WAIT 90 seconds.] Groups post their green answer cards. Teacher cold-calls Amina's group: 'Read your answer card. What evidence from today supports it?' After the group responds, teacher probes: 'What part of this question do we still NOT fully understand?' [WAIT 10 seconds.] Teacher writes any new	Strategy: Driving Question Board (DQB) Revision with Claim Cards. The DQB serves as a public, cumulative record of the class's growing understanding. By physically replacing question cards with evidence-based answer cards, learners experience epistemic progress — they can see what they have explained and what remains open. The new orange questions demonstrate that good scientific explanations generate new questions, mirroring authentic scientific practice.	Observable indicator: Teacher checks that green answer cards reference specific evidence from today's activities (e.g., 'When we moved the mass to the corner, the CG line moved outside the base — that is why the stool topples') rather than restating the question or giving vague answers. Cards that only say 'because of centre of gravity' without referencing the base of support indicate incomplete understanding and flag those groups for re-teaching in Lesson 4.

moments equilibrium centre gravity	moments) and circles them as 'next lesson targets'.	questions on the DQB and says: 'These orange questions are what will drive our next lessons. Notice that we now have the GEOMETRY of toppling — but do we have the NUMBERS? Can we PREDICT the tipping angle? That is our next challenge.'		
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: moments equilibrium centre gravity  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: moments equilibrium centre gravity	Learners return to the cumulative explanatory model they began in Lesson 1 (a labelled diagram of the Jiko stool phenomenon) and revised in Lesson 2 (adding moment arrows). Today, they add a third layer: (a) a dot labelled 'CG' on both the stool-alone and the stool-plus-jerrycan versions; (b) a vertical dashed arrow from the CG pointing downward labelled 'Weight acts here'; (c) the triangular or rectangular outline of the base of support labelled 'Base of Support'; (d) a critical tipping line — a dashed vertical line from CG — coloured red if it falls outside the base and green if inside; and (e) a written caption of no more than three sentences explaining, in the learner's own words, why the stool topples when the jerrycan is at the corner. Groups share their revised models briefly (each group holds up their model; teacher does a rapid visual scan). Teacher asks two targeted questions to the whole class before closing.	Teacher distributes or directs learners to open their Lesson 1 model diagrams. Teacher says: 'Your model has been growing since Lesson 1. Today it gets smarter. I need you to add EXACTLY these four things —' [writes the four additions on the board, numbered]. 'You have 10 minutes. Go.' [WAIT — teacher circulates actively, checking that CG is placed on the combined system, not just the stool alone.] At the 8-minute mark, teacher says: 'Freeze. I am going to ask two questions. Think before you answer.' Question 1: 'If a craftsman in Kericho builds a stool that is taller but narrows the legs — what happens to stability? Tell your neighbour in one sentence.' [WAIT 15 seconds.] Cold-calls Ochieng. Question 2: 'A mama mboga in Gikomba market carries a heavy debe of tomatoes on her head. Why does she walk carefully and keep her back straight?' [WAIT 15 seconds.] Cold-calls Wanjiku. Teacher connects both answers back to the model: 'The physics is IDENTICAL — centre of gravity, base of support, and the tipping line. Your model now explains REAL LIFE.' Teacher closes: 'In Lesson 4, we will use moments calculations to find the EXACT tipping angle. Keep this model — you will need it.'	Strategy: Cumulative Annotated Model Revision. The model-building phase is not a separate activity — it IS the sensemaking product. By requiring learners to add a new conceptual layer (CG, base of support, tipping line) to the same diagram they have been developing since Lesson 1, the model becomes a visible record of conceptual growth. The mama mboga analogy grounds abstract physics in a vivid Kenyan cultural context, reinforcing transfer.	Observable indicator: Teacher's rapid visual scan during the hold-up checks for three non-negotiables: (1) CG dot is on the combined stool-plus-jerrycan system, not at the jerrycan alone; (2) the tipping line (vertical from CG) is drawn and correctly colour-coded red/green based on position relative to the base; (3) the three-sentence caption uses the words 'centre of gravity', 'base of support', and 'topples'. Models missing any of the three are flagged for the teacher to address at the start of Lesson 4 with a quick targeted re-teach.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully transfer the plumb-line CG finding to the stool diagram, or did a significant number still place the CG at the geometric centre regardless of mass distribution? If so, plan a brief 5-minute re-teach at the start of Lesson 4 using a lollipop stick with a lump of ugali clay on one end to make mass distribution visceral. (2) During the Gallery Walk, were learners' sticky-note comments surface-level ('good diagram') or evidence-based ('the CG line goes outside the base here')? If surface-level, introduce a sentence starter scaffold next time: 'This diagram shows _____ because _____.' (3) Did the mama mboga and Kericho craftsman examples resonate? Note which analogies generated the most engagement — these are your cultural anchors for Lesson 4. (4) Review the 6 collected annotated Jiko stool diagrams before Lesson 4: tally how many correctly show CG on the combined system. If fewer than 4 of 6 are correct, open Lesson 4 with a whole-class error analysis of an anonymised incorrect diagram before proceeding to tipping angle calculations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that a plumb-line hanging from different points on an irregular cardboard shape always intersects at the same point (the centre of gravity), and that a tall narrow block tipped when a mass was moved to its edge while a wide low block did not — directly mirroring the jerrycan-on-stool behaviour from the anchoring phenomenon.
What did I learn?	Students learned that every object has a centre of gravity — the single point through which its total weight appears to act — and that a body topples when the vertical line through its centre of gravity passes outside its base of support; they also distinguished between stable, unstable, and neutral equilibrium using this principle.
How does this explain the phenomenon?	Students explained why the Jiko stool topples when the jerrycan is placed at the corner: the jerrycan raises and shifts the combined centre of gravity of the stool-plus-load system until the vertical line through the CG falls outside the triangular base of support formed by the three legs, generating an unbalanced toppling moment with no restoring moment to counteract it.

LESSON 4 (80 minutes): The Principle of Moments: Balancing Forces on a Bao Plank

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate the Principle of Moments experimentally and apply it to solve quantitative equilibrium problems in Kenyan everyday contexts.
Knowledge	By the end of this lesson, learners will be able to: (a) define the moment of a force and state its SI unit (Newton-metre); (b) state the Principle of Moments; (c) calculate clockwise and anti-clockwise moments in given scenarios; (d) apply the Principle of Moments to solve quantitative problems involving balanced beams.
Skills	By the end of this lesson, learners will be able to: (a) plan and conduct a fair experiment using improvised materials (metre rule, pencil pivot, bags of beans, coins) to gather moment data; (b) tabulate and analyse clockwise and anti-clockwise moment values; (c) identify patterns in data and generalise them into the Principle of Moments; (d) use the principle to solve structured problems including a traditional Kenyan see-saw scenario.
Attitudes	By the end of this lesson, learners will: (a) appreciate that systematic data collection leads to reliable scientific principles; (b) demonstrate responsibility by handling shared lab materials carefully; (c) show curiosity in connecting mathematical patterns to the real-world behaviour of the Jiko stool and other balanced objects.
Key Inquiry Question	How do forces determine the stability and equilibrium of objects we encounter every day?
Purpose in Storyline	In Lessons 1–3 learners identified turning forces (moments), located the centre of gravity of irregular objects, and connected the position of the centre of gravity to toppling. This lesson is the quantitative heart of the sequence: learners measure moments directly, discover that balance requires clockwise moments to equal anti-clockwise moments, and use this principle to predict whether a load placement on the Jiko stool will cause toppling — adding precise, calculable meaning to the phenomenon for the first time.
Safety Notes	Ensure the metre rule is stable on the pencil pivot before releasing masses. Bags of beans should be tied securely to prevent spillage. Coins should not be placed near the edge of desks to avoid falling. Learners must not stand on stools or furniture during any demonstration. Remind learners to keep the workspace tidy to prevent trip hazards.

B. LESSON OVERVIEW

Learners begin by reviewing the Driving Question Board (DQB) to identify what is still unexplained about the Jiko stool phenomenon — specifically, WHY position of the load matters quantitatively. Working in groups of four, they set up a metre-rule balance on a pencil pivot (representing the log fulcrum of a Kenyan bao see-saw plank) and systematically vary the mass and distance on each side, recording clockwise and anti-clockwise moments in a structured table. After pooling class data, learners spot the pattern (clockwise moments = anti-clockwise moments at balance) and formally state the Principle of Moments. The teacher then bridges back to the Jiko stool: learners calculate the moment of the jerrycan at different positions and determine the threshold at which toppling occurs. The lesson closes with model revision — groups annotate their cumulative diagrams with moment arms, force arrows, and the balance-condition label — and the DQB is updated to mark newly answered sub-questions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: moments equilibrium principle clockwise  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: moments equilibrium principle clockwise	Learners individually examine a labelled diagram of a bao plank (a 1-metre wooden plank balanced on a log fulcrum at its midpoint) with two Kenyan children sitting at different positions. Child A (mass 40 kg) sits 30 cm from the centre; Child B (mass 20 kg) sits on the opposite side at an unknown distance. Learners write individual predictions: (1) Will the plank be balanced, tip to A's side, or tip to B's side? (2) Where must Child B sit for balance? (3) What rule or pattern might explain your prediction? Learners then share predictions with a partner (Think-Pair) before four predictions are cold-called to the board — teacher records them without evaluation under the heading 'Our Predictions'.	Display the bao plank diagram on the board or printed handout. Say: 'Before we do any measuring today, I want your brain to take a risk. Look at this diagram of Amina and Juma on a bao plank. Predict: will it balance? Where must Juma sit? Write your thinking — numbers welcome, even if you are guessing.' Allow 3 minutes of silent individual writing. After Think-Pair (2 minutes), cold-call exactly 4 students: 'Fatuma — what did you predict and why?', 'Otieno — do you agree or disagree with Fatuma? What is different about your reasoning?' Record predictions verbatim. Say: 'Hold your predictions. By the end of this lesson you will be able to prove mathematically who is right.' DO NOT validate or correct predictions yet. Wait time: 10–12 seconds after each cold-call before prompting.	Think-Pair-Share with Prediction Anchoring — learners activate prior intuition about balance before data collection. Recording predictions publicly creates cognitive commitment and a reference point for the 'explain' phase, making the moment of insight more memorable.	Teacher scans written predictions and listens to pair talk. Observable indicators: (a) At least 60% of learners attempt a numerical estimate for Child B's distance, showing they intuit that both mass and distance matter. (b) Listen for the word 'distance' or 'farther' appearing in reasoning — this signals readiness to formalise. Red flag: if most learners predict balance based on mass alone (ignoring distance), teacher notes this misconception to address explicitly in Phase 3.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: moments equilibrium principle clockwise  READING: No closely matching reading in the ARES library for this activity. Search ARES	Groups of four set up the metre-rule balance experiment: a 30-cm pencil taped to the desk acts as the fulcrum; the metre rule rests on it at the 50 cm mark. Groups hang small bags of dry beans (labelled in grams using a kitchen scale or known coin masses) at measured distances from the pivot on both sides, adjusting until the rule is perfectly horizontal. Each group completes at least 6 trials, varying mass and distance independently. They record: (1) Mass left (g), Distance left from pivot (cm), Clockwise Moment = $F \times d$ (N·cm); (2) Mass right (g), Distance right from pivot (cm), Anti-clockwise Moment = $F \times d$ (N·cm); (3) Status: Balanced / Tips Left / Tips	Distribute printed data table templates. Demonstrate the setup with one group in front of the class (60 seconds max): 'I am placing a 100 g bean bag at 20 cm on the left. Watch — the rule tips left. Now I move a 50 g bag on the right to 40 cm. Watch what happens.' Ask: 'What did you notice? What two things did I change?' (Wait 10 seconds.) Say: 'Your job is to find the mathematical pattern hiding in the numbers. Every group must complete at least 6 trials. Record EVERY result — even the messy ones.' Circulate every 3–4 minutes. At the Data Checkpoint (15 minutes in), say: 'Pause. Show me your moment column. If you	Structured Data Table with Guided Pattern Spotting — the table design forces learners to compute both moments side-by-side, making the equality relationship visually obvious. The mid-activity checkpoint interrupts procedural work to prompt relational thinking, mirroring how scientists pause to interpret during data collection.	Observable indicators: (a) All groups can correctly compute $F \times d$ for at least 4 balanced trials. (b) At the Data Checkpoint, at least 2 groups spontaneously state 'the two moments are equal' or 'both sides give the same number' — this is the target insight. (c) Monitor for unit confusion (mixing cm and m, or grams and Newtons) — if seen in more than 2 groups, pause the class for a 90-second unit clarification. Red flag: a group that always gets 'balanced' by moving the rule rather than adjusting masses/distances is not controlling variables — redirect immediately.

for readings Search terms: moments equilibrium principle clockwise	Right. Groups also test ONE deliberately unbalanced case and note what happens. Data is recorded in a shared class table projected or drawn on the board. Midway, the teacher pauses all groups for a 2-minute 'Data Checkpoint' to verify units and calculation method before groups continue.	multiplied mass in grams by distance in centimetres — good. Now: what do you notice when the rule is balanced? Talk to your group for 90 seconds.' Cold-call 2 groups to share emerging patterns. Prompt if needed: 'Look at the left moment and the right moment in your balanced trials — what relationship do you see?' Wait 12 seconds.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: moments equilibrium principle clockwise  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: moments equilibrium principle clockwise	The class pools data into the board table. Learners examine all balanced trials and, guided by teacher questioning, articulate the pattern in their own words before the teacher offers the formal statement. Learners then copy and annotate the Principle of Moments in their notebooks with a diagram. Next, they solve three structured problems individually: (1) The bao plank problem from Phase 1 — now solved mathematically, predictions are checked. (2) A sufuria of ugali (mass 3 kg) is placed 0.4 m from the pivot of a market weighing plank; what mass of maize grain placed 0.6 m on the other side will balance it? (3) A Jiko stool leg acts as a pivot: if a 20-litre jerrycan (20 kg) is placed 0.25 m from the central pivot point, what restoring moment would a 5 kg counterweight at 0.5 m provide — and is it sufficient to prevent toppling? Learners compare their Phase 1 predictions to their calculated answers and note what they got right/wrong and why.	Point to the class data table. Say: 'Look at every row where the rule was balanced. Stare at the two moment columns. What is always true?' Wait 12 seconds. Cold-call 3 students. After a learner states the equality, say: 'Say that again in one sentence as if you are writing a law of physics.' Write the student's version on the board. Then say: 'Here is how physicists state it formally:' — write verbatim: 'For a body in equilibrium, the sum of clockwise moments about any point equals the sum of anti-clockwise moments about the same point.' Say: 'Underline the word ANY — this is powerful. It does not matter which point you choose.' For problem 3 (Jiko stool), explicitly bridge: 'We are back to our stool. The jerrycan is not on a see-saw — but we can TREAT the tipping edge as the pivot. Let us calculate.' Pause after each problem for 2 minutes of silent individual work before reviewing. Do not solve for learners — ask: 'What is your pivot? What are your two moments? Which is larger?' Cold-call one learner per problem step.	Pattern → Principle → Application (PPA) Sequence — learners move from raw data pattern to formal scientific principle to quantitative application in one lesson. Explicitly returning to the Jiko stool phenomenon in problem 3 closes the predict-observe-explain loop and demonstrates the explanatory power of the Principle of Moments. Prediction checking in Phase 1 creates a personal 'aha moment' for each learner.	Observable indicators: (a) In their notebooks, learners write the Principle of Moments correctly without copying from board within 2 minutes (cover the board briefly to check). (b) At least 75% of learners correctly solve the sufuria-of-ugali problem independently (correct pivot choice, correct $F \times d$ calculation, correct unknown mass). (c) In problem 3, listen for learners correctly identifying the tipping edge as the pivot — this transfers learning to the phenomenon. Red flag: learners who recalculate using weight (in Newtons) rather than mass — affirm this as more precise and note it as a strength, not an error.
Driving Question Board (DQB)	The DQB is displayed. Groups are given 3 minutes to re-read all posted sub-questions from	Say: 'Our DQB has been growing since Lesson 1. Today we have a powerful new tool — the Principle	DQB Audit with Colour-Coded Progress Tracking — green/yellow sticky note protocol makes	Observable indicators: (a) Each group correctly identifies at least one genuinely answered sub-

Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: moments equilibrium principle clockwise  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: moments equilibrium principle clockwise	Lessons 1–3. Each group nominates one sub-question that today's experiment has now answered and writes a one-sentence evidence-based answer on a green sticky note. They also add at least one new sub-question that today's lesson has raised (e.g., 'Does the shape of the base change the pivot point for toppling?' or 'What happens to moments when more than two forces act?') on a yellow sticky note. The teacher facilitates a 4-minute whole-class review: 3 groups share their green notes; 2 new yellow questions are read aloud and posted.	of Moments. Let us audit our board.' Point to specific sub-questions: 'Group C — does today's experiment answer the question about WHY position of the jerrycan matters? Write your answer on green.' After groups write, cold-call 3 groups to read their green notes. For each, ask the class: 'Do we agree this question is now answered? Thumbs up or thumbs sideways.' For yellow new questions, say: 'Good science always opens new doors. What do you still wonder about?' After 2 new questions are posted, say: 'We will chase these in Lessons 5 and 6.' Update the board visually — move answered questions to a 'We Now Know' column.	knowledge growth visible and motivates learners by showing that their questions are driving the unit. Moving cards to 'We Now Know' provides a tangible sense of epistemic progress and reinforces metacognitive awareness of what counts as an answer.	question (not a trivially obvious one) — this shows integration of lesson content with prior questions. (b) New yellow questions are conceptually relevant to moments or stability (not off-topic) — this shows productive disciplinary curiosity. (c) Class thumbs-up agreement on green notes is above 70% — if lower, the teacher notes the disputed answer for follow-up.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: moments equilibrium principle clockwise  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: moments	Each group retrieves their cumulative model diagram (built across Lessons 1–3, showing the Jiko stool, centre of gravity, tipping edge, and weight arrow). They now add: (1) a labelled moment arm (d) from the tipping-edge pivot to the line of action of the jerrycan's weight; (2) a labelled moment arm for any stabilising force (e.g., the stool's own weight acting through the centre of gravity); (3) a 'balance condition' annotation: 'Stool stable IF: anti-clockwise moment > clockwise moment about tipping edge'; (4) a colour-coded moment arrow (clockwise = red, anti-clockwise = blue). Groups display models and do a 90-second gallery walk to view two other groups' diagrams before returning to make one revision based on what they	Distribute model diagrams and coloured pens. Say: 'Your model has shown us WHERE the centre of gravity is and WHAT a tipping edge is. Today we can make it speak mathematics. Add the moment arms — the perpendicular distance from your pivot to the line of action of each force. Label them d_1 and d_2.' Circulate, checking that moment arms are drawn perpendicular to the force arrows (common error: drawing along the beam rather than perpendicular). Correct gently: 'Is that arm perpendicular to the weight arrow? Check with a set square or the corner of your book.' After gallery walk, say: 'You had 90 seconds to be a critic. What is one thing you will improve in your own model?' Cold-call 2 learners. Close by saying: 'Next lesson we examine	Cumulative Model Annotation with Gallery Walk Critique — annotating the shared model embeds the Principle of Moments into the ongoing explanatory framework for the phenomenon, preventing it from being treated as an isolated formula. The gallery walk introduces peer critique as a scientific practice (analogous to peer review), encouraging learners to refine representations.	Observable indicators: (a) Moment arms are drawn correctly (perpendicular to the force arrow, from the pivot) in at least 3 of 4 groups — teacher checks during circulation. (b) The balance condition annotation uses an inequality (anti-clockwise moment vs. clockwise moment), not just an equals sign — this shows understanding that toppling occurs when equality is broken. (c) After the gallery walk, at least 60% of groups make a substantive revision (not just cosmetic) — this shows genuine engagement with peer critique.

equilibrium principle clockwise	noticed.	what happens when multiple forces act — and we will finally be able to fully explain why the wide stool never topples.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully spot the moment equality pattern from their own data before I stated the Principle — or did I give it away too early? If the latter, how can I extend the Data Checkpoint next time to give groups more time to articulate the pattern themselves? (2) Which of the three quantitative problems caused the most difficulty — was it choosing the correct pivot, converting units, or setting up the equation? Plan a targeted 5-minute opener for Lesson 5 to address the most common error. (3) Were learners able to transfer the Principle back to the Jiko stool context in problem 3, or did they treat it as an abstract formula? If transfer was weak, consider adding a physical demonstration with the actual stool in Lesson 5. (4) How well did the DQB audit go — did groups identify genuinely answered questions, or surface-level ones? Review the green sticky notes after class and plan responses to the yellow questions to ensure Lessons 5 and 6 address learner-generated curiosity. (5) Were moment arms drawn correctly on the cumulative model? If perpendicularity was a widespread error, prepare a short visual correction at the start of Lesson 5 before model revision continues.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners balanced a metre-rule on a pencil pivot using bags of beans and coins, recording mass and distance data across 6+ trials and noting whether the rule was balanced or tipping.
What did I learn?	When a ruler (or any beam) is balanced, the clockwise moment (force $\times$ distance) always equals the anti-clockwise moment about the pivot — this is the Principle of Moments. The position of a load relative to the pivot determines whether a turning effect causes toppling or stability.
How does this explain the phenomenon?	The Jiko stool topples when the jerrycan is placed far from the centre because its clockwise moment about the tipping edge exceeds the anti-clockwise moment of the stool's own weight acting through the centre of gravity. The wider stool has a larger anti-clockwise stabilising moment, so the same jerrycan load does not cause toppling — the moments remain balanced.

LESSON 5 (40 minutes): How Do Engineers Keep Things From Falling? — Factors Affecting Stability and Real-Life Applications

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students investigate how lowering the centre of gravity and widening the base of support independently increase the stability of an object, then apply these principles to analyse real engineering contexts including crane design, double-decker buses, and a Nairobi construction-site case study.
Knowledge	Students will be able to: (1) State that stability increases when the centre of gravity is lowered and when the base of support is widened; (2) Explain that an object topples when the vertical line through its centre of gravity falls outside its base of support; (3) Identify engineering strategies that exploit these principles in structures and vehicles.
Skills	Students will be able to: (1) Conduct a fair-test investigation manipulating one variable at a time (height of centre of gravity, width of base) and recording tipping angles; (2) Analyse a case-study document to extract evidence of moment and stability principles; (3) Update a cumulative explanatory model with new causal labels and engineering examples.
Attitudes	Students appreciate that physics principles underlie everyday engineering decisions that protect human lives, and they develop curiosity about careers in civil, mechanical, and transport engineering.
Key Inquiry Question	How do forces determine the stability and equilibrium of objects we encounter every day?
Purpose in Storyline	Lessons 1 through 4 established WHAT toppling is and HOW to classify equilibrium states. Lesson 5 advances the storyline by answering WHY some designs resist toppling and HOW engineers deliberately manipulate centre of gravity and base width to keep structures and vehicles safe — directly resolving the core puzzle of the Toppling Jiko Stool phenomenon and equipping students to write a full scientific explanation in Lesson 6.
Safety Notes	Ensure clay lumps are secured to boxes before tilting to prevent unexpected falling objects. Students should keep feet clear of the table edge during tipping tests. The Nairobi crane case-study card contains accident imagery; preview with sensitivity and remind students the focus is engineering analysis, not graphic content.

B. LESSON OVERVIEW

In a 40-minute lesson students run two back-to-back mini-investigations using tall cardboard boxes, clay, and adjustable cardboard flaps to isolate the effect of (a) lowering the centre of gravity and (b) widening the base of support on stability. They record critical tipping angles and construct causal explanations. A structured case-study card on a documented Nairobi construction-site crane incident is then analysed in pairs using the moment and stability vocabulary built across the unit. The cumulative class model — a labelled diagram of the Jiko stool — is updated with stability-factor labels and an engineering inset sketch. The lesson ends with a DQB review sticky-note that advances the driving question from classification to prediction and design.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown two identical tall cardboard boxes (roughly 30	Display both boxes prominently at the front. Ask: 'Look carefully —	Anchoring prediction activates prior knowledge of centre of	Teacher scans written predictions for evidence that students connect

 VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: moments equilibrium centre gravity  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: moments equilibrium centre gravity	cm tall, 10 cm square base). Box A has a palm-sized lump of clay stuck near its top; Box B has the same clay lump stuck to its inside base. The teacher asks: 'Without touching them, predict which box will topple first when I tilt the table surface slowly — and write down the TWO things about the box you think control that.' Students write individual predictions in their exercise books (2 minutes), then share with a neighbour (think-pair, 1 minute). A quick show-of-hands poll is taken: hands up if you predict Box A falls first. The teacher records the class split on the board without revealing the answer. Students also briefly revisit their Lesson 4 cumulative model sketch and annotate it with a question mark next to any factor they feel is still unexplained.	what is the same about these two boxes, and what is different?' Wait 20 seconds for silent observation before calling on responses. After predictions are recorded, say: 'Keep that prediction — we are about to find out whether your physics reasoning is correct.' Avoid confirming or denying any prediction. Circulate to read written predictions; note any student who mentions centre of gravity or base of support — flag them to share during the Explain phase.	gravity (Lesson 3) and equilibrium states (Lesson 4). The think-pair structure ensures every student commits to a reasoned prediction before observation, preventing passive watching.	centre of gravity height or base width to stability. Students who write only 'the heavier one falls' reveal a misconception about weight vs. turning effect that must be addressed during explanation.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: moments equilibrium centre gravity  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: moments equilibrium centre gravity	INVESTIGATION A — Centre of Gravity Height (10 minutes): Groups of 4 receive: one tall cardboard box (approx. 30 cm x 10 cm x 10 cm), two equal lumps of clay (~100 g each), a protractor, and a ruler. Step 1 — Baseline: Stand the box empty on the desk; tilt it slowly until it just topples; record the critical angle using the protractor. Step 2 — Clay at top: Fix both clay lumps inside at the top of the box with tape; repeat the tilt test; record the critical angle. Step 3 — Clay at base: Move both clay lumps to the inside base; repeat; record. Students tabulate: Configuration Critical tipping angle Toppled easily? (Y/N). INVESTIGATION B — Base of Support Width (8 minutes): Each group receives a second identical empty box with four scored cardboard flaps at its base that	Before Investigation A begins, say: 'Your job is to change ONE thing at a time and keep everything else the same — that is what makes this a fair test. Tell me: what are you keeping the same?' Wait 15 seconds; accept responses. During the investigations, circulate and prompt with: 'Your critical angle went up from 12 degrees to 31 degrees when you moved the clay to the base — what does that tell you about the turning effect needed to topple it?' For groups who finish early, prompt: 'Can you sketch where the centre of gravity is in each configuration — and draw the vertical line through it to show when it crosses the base edge?' During case-study reading, prompt: 'Find the word radius in the card — what does radius mean in terms of the moment formula?' After all observations, call the	Back-to-back controlled investigations enforce the scientific practice of changing one variable at a time, making the causal claims ('lowering centre of gravity increases stability' and 'widening the base increases stability') directly observable rather than told. The case study bridges controlled lab findings to a real high-stakes engineering failure, advancing the storyline toward the driving question.	Teacher checks data tables for correct identification of the independent variable in each investigation. Listening for the phrase 'the angle got bigger' is insufficient — teacher probes: 'Why did a bigger angle mean more stable?' to ensure students link angle to the vertical-line-outside-base criterion from Lesson 4.

	can be folded outward to widen the footprint. Step 1 — Flaps folded in (narrow base, approx. 10 cm x 10 cm): Tilt until topple; record angle. Step 2 — Two flaps out (medium base, approx. 10 cm x 20 cm): Repeat. Step 3 — All four flaps out (wide base, approx. 10 cm x 30 cm): Repeat. Students record results in the same tabular format. CASE STUDY (3 minutes reading): Each pair receives a laminated case-study card titled 'Nairobi Construction Site — Tower Crane Incident, 2015'. The card describes (in student-accessible language) how a tower crane was loaded asymmetrically — a heavy concrete block was lifted and swung beyond the rated radius — shifting the effective centre of gravity outside the crane base, causing the structure to overturn. Two guided questions are printed on the card: (1) Using the word moment, explain why the crane toppled. (2) Suggest TWO design changes that could have prevented the toppling.	class together and ask a group spokesperson to share one surprising result before moving to the Explain phase.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: moments equilibrium centre gravity  READING: No closely matching reading in the ARES library for this activity.	Whole-class sense-making (8 minutes). The teacher draws a quick side-view diagram of the tall box on the board — first with clay at the top, then with clay at the base — marking the centre of gravity (x) and the base of support. Students are invited to come up and draw the vertical line through the centre of gravity at the critical tipping moment in each case. The class discusses: 'Which configuration needs the box to lean further before the vertical line exits the base?' Students verbalise: 'When the centre of gravity is lower, the box must tilt more before it topples — so it is	When drawing the diagram, use a think-aloud: 'I am placing the x for centre of gravity higher up here — now watch what happens to the vertical line as I tilt the box.' Pause 10 seconds after drawing each vertical line and ask: 'Has the line left the base yet? No? So has it toppled?' After students write their two sentences, cold-call three students and ask them to read their sentence aloud. If a student says 'heavier base makes it stable,' redirect: 'Is it the mass that matters, or where the mass is placed — and why?' Reference the Principle of Moments from Lesson 2: 'Remember — moment equals	Diagram annotation by students (not just the teacher) ensures active sense-making. The two-sentence writing task is a low-stakes formative writing move that forces each student to articulate a causal claim. Clustering the case-study suggestions under the same two headings students discovered in the lab closes the loop between investigation and application.	Teacher reads the two written sentences for causal language. Red flag: sentences that say 'lower centre of gravity means the object is heavier at the bottom' — this conflates mass distribution with weight, requiring a follow-up probe. Green flag: sentences that use the vertical-line-through-centre-of-gravity criterion explicitly.

 Search ARES for readings Search terms: moments equilibrium centre gravity	more stable.' The same diagrammatic analysis is done for the wide-base vs. narrow-base box. Students then write a two-sentence causal explanation in their books: Sentence 1 links centre of gravity height to stability; Sentence 2 links base width to stability. The teacher introduces three Kenya-relevant engineering examples using a mini-poster or whiteboard sketches: (1) Double-decker Citi Hoppa buses in Nairobi — heavy engine and battery packs kept low, narrow upper deck; (2) Grain storage silos at the Eldoret cereal board — wide circular base, low centre of mass; (3) Traditional Turkana water-pot carriers — wide-brimmed pot base resting on a head ring, keeping the load centred over the carrier. For the case study, pairs share their two design-change suggestions; the teacher consolidates responses on the board under two headings: 'Lower the centre of gravity' and 'Widen the base of support', reinforcing that engineers use the same two strategies students just discovered in the lab.	force times perpendicular distance. When the centre of gravity shifts outside the base, the weight itself creates a toppling moment. Heavier is not the same as more stable.' Connect back to the anchoring phenomenon explicitly: 'So now we can explain the Jiko stool — which of these two strategies did the wider four-legged stool use?'		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: moments equilibrium centre gravity  READING:	Students receive one green sticky note each. The teacher directs them: 'Write one piece of NEW EVIDENCE from today that helps answer the driving question — Why do some objects stay balanced while others topple?' Students write individually (2 minutes), then come up and place their note on the DQB in the section labelled Lesson 5 Evidence. Two or three notes are read aloud. The teacher also draws attention to any Lesson 1 or Lesson 2 sticky-note questions that can now be fully answered	Say: 'Today you moved from describing stability to explaining it and now to applying it. Your green note should show that shift — do not write what you did; write what it proves.' After notes are posted, ask: 'Which Lesson 1 question on the board can someone now answer using today and all our previous lessons?' Wait 20 seconds. Affirm correct answers and mark answered questions with a tick. Leave unanswered questions visible for Lesson 6.	The DQB move makes the progressive accumulation of evidence visible and narrative — students see their own questions getting answered over time, reinforcing the storyline coherence and metacognitive awareness of their own learning trajectory.	Teacher reads green sticky notes for evidence of transfer: does the student cite a specific observation (angle, clay position, flap width) or only a vague claim? Vague notes signal that the student has not yet connected observation to explanation and will need targeted support in Lesson 6.

<i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: moments equilibrium centre gravity	(e.g., 'Does a wider stool always win?' — YES, because a wider base means the vertical line through the centre of gravity must travel further before leaving the base of support). The teacher posts one new orange prompt card: 'Can we now PREDICT, before building anything, whether a design will topple or stay stable — and can we calculate it?' This bridges to Lesson 6 consolidation.			
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: moments equilibrium centre gravity  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: moments equilibrium centre gravity	Students open their cumulative model — the annotated Jiko stool diagram maintained since Lesson 1. Working in pairs, they add two new layers: (1) Stability-factor labels — two arrows or callout boxes, one pointing to the stool base with the label 'Wider base — vertical line through centre of gravity must travel further to exit base — MORE STABLE', and one pointing to the seat surface with the label 'Load kept low — centre of gravity stays low — MORE STABLE'; (2) Engineering example inset — a small sketch in the corner of either a Citi Hoppa double-decker bus or a construction crane, labelled with the two stability strategies. Students also update the Stool-vs-Jerrycan scenario: they annotate where the centre of gravity shifts when the jerrycan is placed on the corner and add a note explaining why the narrow stool topples but the wide stool does not. Pairs share their updated models with an adjacent pair and offer one piece of specific feedback using the sentence starter: 'Your model clearly shows _____ but could add _____ to make the causal chain clearer.'	Circulate and check that students are adding CAUSAL labels (not just descriptive labels). Prompt: 'Your arrow points to the base — what does the wider base actually do to the vertical line through the centre of gravity? Can you add those words?' For the engineering inset, ask: 'Which of the two strategies does a double-decker bus use — lowering centre of gravity, widening base, or both — and where exactly?' During peer feedback, listen for whether feedback is specific and physics-based. If it is vague (e.g., 'it looks good'), prompt: 'Can you find one place where the causal explanation is missing — where it says what happened but not why?' Before the lesson ends, ask one pair to hold up their model and describe one update to the class in 30 seconds.	Cumulative model building is the core epistemic practice of the unit — each lesson deposits a new explanatory layer, so by Lesson 6 the model is a complete, student-constructed explanation of the driving question. Peer feedback with a sentence starter scaffold ensures feedback is substantive rather than social.	Teacher photographs three to four models (with student permission) to project at the start of Lesson 6 as anchor examples. Assessment criterion: Does the model show the vertical-line-through-centre-of-gravity mechanism explicitly, or only name the two factors without explaining the mechanism? The latter is the key gap to close in Lesson 6.

D. TEACHER REFLECTION

After the lesson, ask: (1) Did students successfully isolate one variable at a time in both investigations, or did groups inadvertently change two variables simultaneously? If yes, plan a brief fair-test review at the start of Lesson 6. (2) Were students able to articulate the vertical-line-through-centre-of-gravity mechanism in their written sentences, or did they revert to describing observations? If the latter, consider projecting two contrasting student sentences at the start of Lesson 6 (anonymised) and asking the class to identify which is causal and which is merely descriptive. (3) Did the case-study card generate genuine engagement with the engineering context, or did it feel like a worksheet add-on? If engagement was low, in Lesson 6 invite students to suggest a local Kenyan infrastructure project (e.g., the Nairobi Expressway bridge piers, the Standard Gauge Railway viaducts) and ask how the same principles apply. (4) Check cumulative models photographed — are the stability-factor labels causal or merely descriptive? Use these as diagnostic data for targeted support in Lesson 6 consolidation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When clay was added to the TOP of the tall box, the critical tipping angle was small (the box toppled easily). When clay was moved to the BASE, the critical tipping angle was much larger (harder to topple). When cardboard flaps widened the base, the critical tipping angle increased further. In the case study, the crane toppled when its load was swung beyond the rated radius, shifting the effective centre of gravity outside the base.
What did I learn?	Stability increases when the centre of gravity is lowered and when the base of support is widened. An object topples when the vertical line through its centre of gravity falls outside the base of support. Engineers apply both strategies deliberately in crane design, double-decker bus geometry, silo construction, and load distribution.
How does this explain the phenomenon?	In the Jiko stool phenomenon, the narrow three-legged stool toppled when the jerrycan was placed at the corner because the combined centre of gravity of stool-plus-load shifted far to one side, and the tall load kept the centre of gravity high — so only a small nudge moved the vertical line outside the narrow base. The wide four-legged stool survived because its broad base meant the vertical line had to travel much further before exiting the base of support, even with the jerrycan at the corner.

LESSON 6 (40 minutes): Can We Explain and Predict Toppling? — Consolidation, Model Finalisation, and DQB Review

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all learning from the sub-strand by using the completed cumulative model to fully explain the anchoring phenomenon, answer every Driving Question Board question with evidence, and demonstrate mastery through a summative problem set.
Knowledge	Students will be able to: (1) state and apply the formula $\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$; (2) state the Principle of Moments and apply it to predict balance or toppling; (3) define centre of gravity and explain how its position relative to the base of support determines the state of equilibrium; (4) distinguish stable, unstable, and neutral equilibrium with physical reasoning; (5) explain how lowering the centre of gravity and widening the base of support each increase stability.
Skills	Constructing and interpreting scientific models; quantitative problem solving involving moments; peer assessment using a co-constructed rubric; evidence-based scientific explanation; group presentation and scientific communication.
Attitudes	Appreciation of the relevance of Physics in everyday Kenyan life; confidence in using physical principles to predict real-world outcomes; collaborative respect during peer assessment; curiosity about engineering and design careers.
Key Inquiry Question	How do forces determine the stability and equilibrium of objects we encounter every day? — addressed fully by synthesising moment calculations, centre of gravity location, and stability principles to explain and predict toppling of the Jiko stool and related objects.
Purpose in Storyline	This is the final lesson of the storyline. Students bring together every concept built across Lessons 1 to 5 — turning effects, the Principle of Moments, centre of gravity, states of equilibrium, and stability factors — into one coherent explanation of the anchoring phenomenon. The Driving Question Board is closed with evidence-based answers, and a summative task confirms individual mastery before the sub-strand ends.
Safety Notes	No hazardous chemicals. If physical props (stool, jerrycan) are used for demonstration, ensure the demonstration area is clear of students. Supervise any stacking of objects. Students with physical disabilities should be seated with clear sightlines to demonstration and presentation areas.

B. LESSON OVERVIEW

In this 40-minute consolidation lesson students begin by revisiting the anchoring phenomenon — the Jiko stool and the jerrycan — and individually predict, using precise physical language, whether a modified scenario will topple or remain stable. They then observe a teacher-led re-enactment or short video clip of the phenomenon and compare outcomes with their predictions. In the Explain phase each group presents their finalised cumulative model, justifying every annotation with evidence from Lessons 1 to 5 and receiving structured peer feedback. The Driving Question Board is reviewed collectively: every student question posted across the unit is answered aloud with evidence. The lesson closes with an individual summative problem set — including a jerrycan-on-stool moment calculation — that assesses all five learning outcomes. Career connections (structural engineering, physiotherapy, aircraft loading, crane operation) are woven throughout. The lesson is explicitly tied to KICD key inquiry questions about the relevance of Physics in daily life and the role of forces in stability.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Introduction to torque <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Torque (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Students are shown a projected image of two stools side by side: the original three-legged Jiko stool (tall, narrow base) and the wider four-legged stool from the anchoring phenomenon. A new scenario is added — a maize sack (approximately 90 kg) placed at the far left corner of the three-legged stool, and a second scenario where the same sack is placed at the centre of the wide stool. Students individually write a prediction card (provided index card) answering three prompts: (1) Will each arrangement topple or remain stable? (2) What is the deciding physical condition? (3) If I lower the maize sack to the ground level (i.e., lower centre of gravity of the loaded system) would your answer change — and why? Students have 4 minutes of silent individual thinking. They then share with their partner using a Think-Pair routine. Predictions are NOT yet confirmed — they are posted on a designated Prediction Wall strip.	Teacher projects the two-stool image and reads each prompt aloud clearly. Teacher says: 'You have spent five lessons building the tools to answer this. Do not guess — reason. Use the words moment, centre of gravity, base of support, and toppling condition in your written prediction.' WAIT TIME: Allow 3 full minutes of silent individual writing before any partner talk begins — teacher times this visibly on a phone or watch and does not speak during the silence. Teacher circulates and reads over shoulders without commenting. After partner sharing, teacher calls on two pairs — one who predicted toppling and one who predicted stability for the same scenario — and says: 'Interesting. We have two different predictions for the same stool. Let us observe and find out who the evidence supports.' Teacher does NOT reveal the answer yet.	Activating prior knowledge through structured prediction prompts; Think-Pair routine surfaces misconceptions before the lesson proceeds; posting predictions publicly creates accountability for revision after observation.	Teacher reads prediction cards over students shoulders during silent time. Target check: do students use the phrase vertical line through the centre of gravity falls outside the base of support rather than vague language like it is heavy or it will fall? Cards with vague language are flagged mentally for targeted questioning during the Explain phase. No marks are given at this stage.
Observe Phase  VIDEO: Introduction to torque - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Torque Practice <i>Related topic: not an exact match</i>	Teacher or a student volunteer performs a live demonstration at the front: (A) places a filled 5-litre container (or equivalent improvised mass) at the far corner of the three-legged stool model (can be a small wooden stool replica or a cardboard model built in a previous lesson) and gives a gentle lateral nudge — students observe toppling; (B) places the same container at the centre of the wide flat model — students observe it rocks and recovers; (C) places the container at the corner of the wide flat model — students observe it is close to the tipping threshold. Students record	Before the demonstration teacher says: 'Watch the moment the stool begins to tilt. Ask yourself — where is the vertical line through the centre of gravity at that exact instant?' WAIT TIME: After the toppling demonstration, teacher pauses for 20 seconds of silence and says nothing — allowing students to process what they saw before asking: 'Turn to your prediction card. Without erasing, write the word REVISED if your prediction needs updating and explain in one sentence why the evidence changed your thinking.' Teacher then asks: 'Who had a partial match — you got the	Live demonstration anchors abstract concepts in direct sensory experience; the three-plus-one column table scaffolds disciplined observation rather than casual watching; the animation bridges observation and theory by making the invisible (centre of gravity line) visible; public prediction revision normalises science as an iterative process.	Teacher scans REVISED annotations. If students write only 'because the centre of gravity moved' without specifying relative to the base of support, teacher makes a mental note and addresses this gap explicitly in the Explain phase introduction. Teacher asks one targeted question to the whole class: 'Give me the exact tipping condition in one sentence using all four of these words — vertical line, centre of gravity, base of support, outside.' Cold calls two students after 10 seconds of wait time.

for this lesson. Source: CK-12 Search ARES for similar readings	observations in their notebooks in a three-column table: Scenario What I observed (exactly) Matches my prediction (yes/partial/no). Teacher then plays a 90-second projected animation or hand-drawn diagram sequence showing how the vertical line through the combined centre of gravity shifts as the load moves outward — crossing or not crossing the base of support boundary. Students annotate their observation table with a fourth column: Physical reason (using unit vocabulary). The observation phase takes approximately 7 minutes total.	outcome right but for the wrong reason? Raise your hand.' Teacher acknowledges these students: 'That is excellent scientific honesty. The goal is to match both the outcome AND the reasoning.' Teacher uses the animation to ask: 'At which exact moment does the stool become committed to toppling — before the nudge, during, or after?' WAIT TIME: 15 seconds before taking responses.		
Explain Phase  VIDEO: Introduction to torque <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Torque <i>Related topic: not an exact match for this lesson.</i> Source: PhET Search ARES for similar readings	Each group (4 to 5 students) displays their finalised cumulative model — the annotated diagram built progressively across all six lessons. Each model should show: force arrows and moment arms (Lessons 1–2); centre of gravity marker and stability state annotations (Lesson 3); clockwise and anticlockwise moment labels with the balance condition (Lesson 4); stability factor labels and an engineering inset (Lesson 5); and now in Lesson 6 a final written explanation strip attached to the bottom of the model answering the driving question. Groups take turns (approximately 2 minutes per group) to present one chosen annotation and explain it to the class — they do not read it but explain it in their own words. Peers use a co-constructed peer assessment rubric (prepared on a half-sheet) with four criteria: (1) Scientific accuracy of explanation — is moment defined and applied correctly? (2) Use of evidence — is each claim linked to an	Teacher introduces the Explain phase by saying: 'Your model is not a decoration. It is a scientific argument. Every arrow, every label, every annotation must be something you can defend with evidence from our experiments. If a peer asks why, you must be able to answer.' During presentations teacher listens for three common errors and addresses them if they occur: (1) Confusing moment arm with total distance — teacher draws a perpendicular on the board and asks the class: 'Is this the right distance — yes or no?' WAIT TIME 10 seconds. (2) Saying the centre of gravity is always at the geometric centre — teacher holds up the cut-out Kenya map from Lesson 3 and asks: 'Is the centre of gravity of this shape at its geometric centre? What does your experiment from Lesson 3 tell you?' (3) Defining stable equilibrium only as returning to position without mentioning the vertical line condition — teacher prompts: 'What must be true about	Group model presentation externalises thinking and requires students to translate private understanding into public scientific language; co-constructed peer rubric develops metacognitive awareness of quality scientific explanation; collective DQB review provides narrative closure to the unit storyline; scribing the consensus answer models how scientists consolidate evidence into a shared claim.	Teacher uses the peer rubric scores to identify groups scoring 1 on scientific accuracy. These groups are pulled aside during the summative problem set for a brief targeted re-teaching exchange. Teacher listens for whether students can connect each DQB question to a specific lesson and a specific piece of evidence — inability to do so reveals surface rather than deep understanding. Teacher notes which DQB questions remain unresolved for the class and plans a brief post-lesson follow-up if needed.

	observation or data from the unit? (3) Clarity of the model — can a newcomer understand it? (4) Connection to the anchoring phenomenon — does the explanation refer back to the Jiko stool? Each peer group awards 1 (developing), 2 (achieving), or 3 (exceeding) for each criterion and writes one warm (strength) and one wonder (question or suggestion). After presentations, the class collectively answers the driving question aloud: teacher scribes a class consensus answer on the board which students copy into their notebooks.	the vertical line for that return to happen?' After all presentations teacher scribes the class consensus answer to the driving question: 'Objects topple when the vertical line through the combined centre of gravity falls outside the base of support, producing an unbalanced moment that rotates the object away from equilibrium. Stability is increased by lowering the centre of gravity and widening the base of support, as engineers deliberately design in cranes, buses, and buildings.' Teacher then reviews the Driving Question Board: reads each posted question aloud, asks the class for the evidence-based answer, and physically moves or ticks each card. Teacher says for each resolved question: 'Which lesson gave us the evidence to answer this?' For any question the class cannot yet answer fully, teacher acknowledges it openly: 'That is a question that takes us beyond this sub-strand — it is a great starting point for further study.'		
Driving Question Board (DQB) Creation  VIDEO: Introduction to torque - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Torque	Although the DQB was created progressively from Lesson 1, this phase in Lesson 6 serves as the formal closure of the board. Students first do a silent gallery walk (2 minutes) around the room to re-read all cards posted since Lesson 1. Each student places a small green dot sticker on the card whose question they feel the unit answered most completely, and a small orange dot on any card they feel is still not fully answered for them personally. The class then gathers and the teacher leads a 4-minute structured discussion: three green-dot questions are read aloud and answered collectively	Teacher facilitates the gallery walk by saying: 'Read every card as if you are a scientist checking whether your community of learners has answered the question you posted weeks ago. Be honest with your dots.' WAIT TIME: The full 2 minutes of gallery walk are silent — teacher enforces this and does not narrate. During the structured discussion teacher says for each green-dot question: 'Before I ask for the answer, tell me — which experiment gave us the evidence? Name the lesson.' For orange-dot cards teacher says: 'If you still have this question, you are thinking like a	Green and orange dot voting makes collective and individual knowledge gaps visible without shame; career connection cards bridge Physics content to student identity and future aspirations; the teacher modelling a career card reduces hierarchy and builds classroom community; naming future learning destinations respects students intellectual curiosity without leaving questions unresolved.	Teacher photographs the DQB with orange dots for personal records — clusters of orange dots on specific questions signal topics needing re-teaching or additional resources in future iterations of the unit. Teacher listens to career connection reasoning: statements that say only 'engineers use Physics' without a specific moment or stability link indicate surface-level career awareness and are gently probed with 'How specifically does the concept of toppling show up in that job?'

(Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	with evidence; any orange-dot questions are discussed — if resolvable with unit knowledge, a student volunteer answers; if genuinely beyond the sub-strand, the teacher names the future topic (e.g., torque in rotational dynamics at A-level, structural engineering in technical training). Students add one final card to the DQB: a Career Connection card. Each student writes on a new card: 'A career that uses moment and stability principles is [CAREER] because [REASON LINKED TO PHYSICS].' Examples suggested by teacher if needed: physiotherapist (body centre of gravity in rehabilitation), structural engineer (building foundation design), aircraft load controller (load distribution for stable flight), Nairobi crane operator (safe load moments). Cards are posted and teacher reads three aloud, affirming the Physics-career link explicitly.	scientist who knows the boundary of current knowledge. Here is where this question leads next in your Physics journey.' For Career Connection cards teacher says: 'Physics is not only in laboratories. Every matatu driver who loads passengers evenly, every mama mboga who balances a tray on her head, every JKIA aircraft engineer who checks load sheets — they are all applying what you have learned. Your future career will almost certainly touch these ideas.' Teacher posts own Career Connection card: 'A Physics teacher uses the principle of moments every time they decide where to stand relative to a stubborn classroom door hinge.' This models humour and self-inclusion.		
Model Building Phase  VIDEO: Introduction to torque <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Torque Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12	Students complete the final update to their cumulative model by attaching a Explanation Strip — a written paragraph (minimum 5 sentences) at the bottom of their diagram. The strip must: (1) state when an object will topple (toppling condition); (2) reference the anchoring phenomenon specifically (the Jiko stool and the jerrycan); (3) use the terms moment, centre of gravity, base of support, stable equilibrium, and perpendicular distance each at least once; (4) include one quantitative example — students calculate the moment of the 20-litre jerrycan (mass 20 kg, $g = 10 \text{ N/kg}$, pivoting about the nearest stool leg, perpendicular distance	Teacher introduces the Explanation Strip task by saying: 'This strip is the scientific legacy of your model. If someone picks up your diagram in ten years, this paragraph must be enough for them to understand the physics of toppling without asking you a single question. Write it for that future reader.' Teacher circulates during individual summative work, ensuring no collaboration. Teacher reads over shoulders for Question 5 specifically — if students subtract moments rather than compare them, teacher points to the student work and says quietly: 'Which moment is larger? What does a larger net moment tell you about the direction of rotation?' —	The Explanation Strip externalises and consolidates integrated understanding in written scientific language; the quantitative example in the strip requires students to bridge conceptual and mathematical understanding; the summative problem set provides individual accountability across all five learning outcomes; the Reflection Prompt develops metacognitive awareness of conceptual change; the teacher closing call-back to the anchoring phenomenon gives the unit narrative a satisfying and memorable closure.	Teacher collects summative problem sets at the end of the lesson. Marking focus: Question 5c is the highest-order item — students must correctly compare 80 Nm (jerrycan moment: 200×0.4) with 3 Nm (stool countermoment: 30×0.1) and conclude toppling occurs because the clockwise moment is greater. Students who reach the correct numerical answer but justify using only qualitative language (it is heavy) score partial credit and are identified for feedback. Reflection prompts are scanned for evidence of genuine conceptual change — prompts that restate the new idea in the same vague language as the old idea (now I understand

Search ARES for similar readings	estimated as 0.3 m) and state whether this moment is balanced or unbalanced given the counteracting moment of the stool weight (stool mass 3 kg, centre of gravity 0.15 m from the same leg). Students then complete the individual summative problem set (5 questions printed on a half-sheet). Question 1: Define moment of a force and state its SI unit. Question 2: A student pushes a school gate with a force of 40 N at 0.8 m from the hinge. Calculate the moment. Question 3: A see-saw has a child of weight 300 N sitting 1.2 m from the pivot. Where must a second child of weight 400 N sit for the see-saw to balance? Question 4: Explain why a double-decker bus is designed with a heavy lower deck and a lighter upper deck. Use the terms centre of gravity and base of support. Question 5 (the jerrycan-on-stool problem): A 20-litre jerrycan of water (weight 200 N) is placed 0.4 m from the pivot leg of a three-legged stool. The stool weighs 30 N and its centre of gravity is 0.1 m on the other side of the pivot leg. (a) Calculate the clockwise moment of the jerrycan. (b) Calculate the anticlockwise moment of the stool weight. (c) State whether the stool topples or remains stable and justify your answer using the Principle of Moments. After completing the problem set individually, students write a personal Reflection Prompt in their notebooks: 'Before this unit I thought balance and toppling were about [OLD IDEA]. Now I understand that [NEW UNDERSTANDING] because [EVIDENCE FROM AN	without giving the answer. WAIT TIME: After distributing the summative sheet, teacher allows 30 seconds of silent reading before students begin writing — this reduces anxiety and allows students to plan their approach. At the 5-minute mark teacher announces: 'You should be on Question 3 or 4 by now — manage your time.' At the end of the summative, teacher says: 'Write your reflection prompt. Be honest. The most powerful learning happens when you name exactly what changed in your thinking and exactly why.' Teacher closes the lesson by returning to the anchoring phenomenon one final time: 'The Jiko stool that puzzled you in Lesson 1 — you can now explain it precisely, calculate its moments, and redesign it to be safer. That is what Physics gives you: not just the answer, but the power to predict and control.' Teacher announces the next sub-strand topic briefly to maintain continuity.		balance) indicate surface learning and trigger a follow-up conversation. Teacher records results in the continuous assessment log against each of the five knowledge outcomes in the SLO.
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	EXPERIMENT WE DID].'			
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Were students able to use the toppling condition — vertical line through centre of gravity falling outside the base of support — precisely and consistently, or did vague language persist despite five lessons of building? If vague language persisted, what scaffolding would have helped earlier in the unit? (2) Did the DQB orange dots cluster on any one concept — for example, the distinction between moment arm and total distance, or the location of centre of gravity in irregular objects? If so, which lesson in the sequence needs to be strengthened in the next iteration? (3) Did the summative problem set reveal a gap between students who can perform moment calculations and students who can interpret what those calculations mean for stability? If so, how can quantitative and conceptual tasks be better integrated in Lessons 1 to 5? (4) Were career connections made authentically — did students name specific Physics principles in their career cards — or were the connections generic? What additional real-world examples from the Kenyan context (Mombasa port crane operators, JKIA load planners, Nairobi high-rise structural engineers) would deepen the relevance? (5) Did the peer assessment rubric generate genuinely useful feedback or did peers inflate each other scores? Would anonymous peer assessment or a moderated sample have been more productive? (6) Did the lesson fit within 40 minutes? If not, which phase ran over and what adjustment is recommended — for example, could the gallery walk be shortened to 90 seconds or the summative reduced to 4 questions without losing coverage of all learning outcomes?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the demonstration I observed that the three-legged stool with the jerrycan placed at its far corner toppled when nudged, while the wider four-legged stool with the same load at its corner only rocked and returned to its upright position. I also observed from the animation that the vertical line through the centre of gravity crossed outside the base of support at the exact moment the narrower stool committed to toppling.
What did I learn?	I learned that an object topples when the vertical line through its combined centre of gravity falls outside the base of support, creating an unbalanced moment that rotates the object away from equilibrium. I also learned that stability is increased by lowering the centre of gravity and by widening the base of support, and that engineers deliberately use both strategies in structures such as cranes, double-decker buses, and grain silos. I can now calculate moments and use the Principle of Moments to predict whether a loaded object will topple or remain balanced.
How does this explain the phenomenon?	The Jiko stool topples because placing the heavy jerrycan at its far corner shifts the combined centre of gravity of the stool-plus-load system outward. When the vertical line through that combined centre of gravity moves beyond the edge of the base of support — the triangle formed by the three legs — the clockwise moment of the load about the nearest leg is greater than the anticlockwise moment of the rest of the stool. This net unbalanced moment rotates the stool until it falls. The wider stool does not topple because its larger base of support keeps the vertical line through the centre of gravity inside the triangle even when the load is near the edge, so the net moment remains balanced or self-correcting. The same physics explains why engineers lower the centre of gravity and widen the base in every structure that must resist toppling forces.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.